A Timing-Only Receiver for OFDM through Dispersive Time-Modulated Reconfigurable Intelligent Surfaces

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Abstract

A time-modulated reconfigurable intelligent surface (TM-RIS) switches its elements several times within each OFDM symbol. The switching creates harmonics that couple every received subcarrier to many transmitted ones, and when the elements are frequency dispersive, as real varactor cells are, the coupling matrix is neither diagonal nor Toeplitz. Existing TM-RIS systems therefore treat the coupling as interference, which caps the rate of the legitimate receiver at the signal-to-harmonic-interference ratio, and a recent physically consistent model of the dispersive TM-RIS channel leaves receiver design open. This paper builds that receiver around one structural fact: grouping the harmonic expansion by switching state separates the coupling into Toeplitz factors fixed by the switching timing, which the receiver knows, and per-state input responses that absorb both the propagation and the element dispersion, which it does not. Every receiver block follows from this separation: a pilot-based low-rank estimator that needs no element model; a blind estimator of the RIS clock offset and of the delay reference, based on the observation that an offset is exactly an opposite delay of the two hops, which makes a wrong offset overfit the pilots; a two-timescale extension that learns the static base-station–RIS side once; warm-started updates; and an exact matrix-free equalizer. In an extensive evaluation with the published cell of the anchor model, physical and 3GPP channels, mobility, hardware impairments and coding, the receiver is within 0.1 bit/s/Hz of perfect channel knowledge with two pilot symbols on a shared-path channel and within 0.2 with four to eight on 3GPP TDL-C at 20 dB; in a pre-registered comparison in which every baseline gets a robust equalizer, it beats the best baseline on every channel by 0.3–0.4 bit/s/Hz at 20 dB and 1.1–2.2 at 30 dB, and with two pilot symbols exceeds a full-harmonic basis-expansion estimator given 32, decodes coded blocks within 0.14 dB of perfect knowledge where the conventional receiver never decodes, absorbs an unknown clock offset blindly with no rate loss against known timing, and keeps 3.9–5.7 bit/s/Hz net of pilot overhead on 3GPP TDL channels from walking speed to 120 km/h. All limits found, including failed attempts, are reported.

Index Terms

Reconfigurable intelligent surface, time modulation, space-time coding, OFDM, intercarrier interference, channel estimation, synchronization, equalization.

I. Introduction

A. Context and problem

A reconfigurable intelligent surface (RIS) shapes the radio channel with many passive, electronically switched elements [2], [3]. Most RIS work keeps the configuration fixed during an OFDM symbol and changes it, if at all, between symbols. A *time-modulated* RIS (TM-RIS), also called a space-time-coding metasurface, instead switches each element several times within the symbol [4]. The reflected signal then contains harmonics of the switching frequency, and the surface can steer, scramble or split energy across frequency as well as space. OFDM-aligned time modulation has been proposed for waveform security and directional modulation [5], [6], [9] and for joint sensing and security [7].

For the user that the surface serves, the same harmonics are a problem. A harmonic at h times the subcarrier spacing moves energy from transmitted subcarrier m to received subcarrier $m + h$, so the frequency-domain channel matrix is no longer diagonal. Iudice *et al.* [1] recently derived a physically consistent model of this channel in which each element has a frequency-dispersive (finite-memory) response, as a real varactor cell does. They showed that a generalized cyclic prefix removes inter-symbol interference but that the coupling between subcarriers remains for any prefix length, and that with dispersive elements the coupling matrix is not even Toeplitz. Receiver design was explicitly left open.

How large is the problem? For a two-state element with ideal reflection ± 1 and duty cycle D (fraction of the period in state 1), Parseval's theorem gives the fraction of reflected energy that stays on the original subcarrier as $(2D-1)^2$ (Appendix A). At $D = 0.75$ only 25% stays; the remaining 75% is spread over the other harmonics. A conventional one-tap OFDM receiver treats that 75% as interference, so its signal-to-interference ratio is at most $(2D-1)^2/(1-(2D-1)^2) = -4.8\,\text{dB}$, and its rate saturates however high the SNR. In our evaluation the one-tap receiver delivers 0.42 bit/s/Hz at both 20 and 30 dB, against 6.6 and 9.9 bit/s/Hz with perfect channel knowledge and joint equalization (Section VII-D).

Time modulation does not, by itself, give the user more rate: for a single user, an optimized *static* RIS delivers more (Section VII-O). A TM-RIS is deployed for its other functions, such as directional security, harmonic steering of several users or sensing, and the question this paper answers is how much of the resulting loss the legitimate receiver can win back. Recovering this rate requires the full coupling matrix at the receiver. The matrix has up to M^2 entries ($M = 3276$ subcarriers in a 100 MHz 5G NR carrier) and depends on the propagation on both hops, on the switching pattern, and on the element

response. A generic estimator of a non-diagonal OFDM channel, such as the basis-expansion model used for high-mobility channels [11], does not know where the coupling comes from. It must estimate every harmonic separately. Because rectangular switching leaves a slowly decaying $1/h^2$ energy tail, it either truncates (and then misses 3.4% of the energy at $|h| \leq 12$, which caps its accuracy near $-15\,\mathrm{dB}$) or keeps all harmonics (and then needs many pilots).

B. Central idea

The receiver of a TM-RIS link is in an unusual position: it knows *how* the channel varies inside the symbol, because the variation is the switching pattern, and the switching pattern is set by the network. What it does not know is the propagation and the exact element response. This paper turns that observation into an exact structural statement and builds the entire receiver on it.

Grouping the harmonic expansion of [1] by element *state* rather than by harmonic gives (Proposition 1)

$$\boldsymbol{H} = \sum_{q=1}^{Q} \sum_{s=1}^{2} \underbrace{\boldsymbol{T}_{q,s}}_{\text{timing only}} \circ \big(\underbrace{\boldsymbol{a}_q^{\mathrm{d}}}_{\text{user hop}} \big(\underbrace{\boldsymbol{\Gamma}_s \odot \boldsymbol{a}_q^{\mathrm{s}}}_{\text{BS hop and dispersion}} \big)^{\top} \big),$$

where each $\boldsymbol{T}_{q,s}$ is a Toeplitz matrix fixed by the switching timing alone. The algebra is a regrouping of [1, eq. (5)], and we credit it as such. What is new is what the regrouping implies for a receiver. Each consequence becomes one block of the receiver:

1) The element dispersion multiplies only the input side, together with the unknown base-station (BS)–RIS channel, so it can be absorbed into an unknown per-state input response. *The receiver needs no element model* (Section V-B). Assuming the ideal ± 1 element of earlier work costs 1.4/6.6 dB at 20/30 dB; the timing-only receiver loses 0.08/0.11 dB.
2) All elements see the same propagation paths, so the unknown factors live in a low-dimensional delay subspace shared by all elements and both states. The unknowns fall from up to M^2 to about $3Lr$ (L delay functions, rank r), and two pilot symbols suffice for a sparse channel (four to eight for 3GPP TDL-C at 100 ns).
3) A clock offset between the RIS and the receiver changes $\boldsymbol{T}_{q,s}$ to $\boldsymbol{D}\boldsymbol{T}_{q,s}\boldsymbol{D}^{*}$, which is *exactly* a delay of the user hop and an advance of the BS hop. This explains why a pilot-residual timing search fails at high SNR (the delays absorb the offset), and why the Bayesian evidence, which also asks how probable the absorbing delays are, recovers it (Section V-C).
4) The BS–RIS side does not move, so it can be learned once over many frames, and each new frame estimates only the user side (Section V-D).
5) Each $\boldsymbol{T}_{q,s}\boldsymbol{x}$ is a linear convolution, so products with $\boldsymbol{H}$ cost $\mathcal{O}(QM \log M)$ and the MMSE equalizer can run matrix-free (Section V-F).

6) The $1/h$ amplitude envelope of $\boldsymbol{T}_{q,s}$ explains, quantitatively, why truncated estimators and banded equalizers fail on this channel (Corollary 3).

C. Contributions

- **C1, receiver structure.** To our knowledge, the first pilot-based receiver for OFDM through a (dispersive) TM-RIS: pilot-based estimation of the complete coupling matrix and joint equalization, organized around the timing/propagation separation above (Fig. 1, Algorithms 1–5).
- **C2, timing-only estimation.** Absorbing the dispersion into the unknown input response, so that no element model, calibration or resonance knowledge is needed, and a low-rank shared-subspace estimator that covers identical, cyclically shifted and arbitrary per-element switching patterns in one form.
- **C3, blind clock-offset recovery.** The opposite-delay identity for a TM-RIS clock offset and its physical meaning (bulk hop delays cancel if the RIS follows the incident signal; the BS–RIS delay becomes an offset if it follows the BS clock), the resulting overfitting mechanism of residual-based timing, and an evidence-based estimator of the offset and the delay reference that absorbs the offset from two pilot symbols with no rate loss against known timing. (Because of the identity, the offset is defined only relative to the reference of the delay prior; the meaningful metric is the rate loss.)
- **C4, exhaustive evaluation.** An evaluation against the strongest baselines we could build (full-harmonic basis expansion given $16\times$ more pilots, sparse recovery, unstructured least squares, literature equalizers given the true channel), on physical and 3GPP channels, with coding, mobility and pilot overhead, hardware impairments, timing errors, multiple panels and interference, and the complexity in operations and measured time. Pass/fail criteria were fixed before each experiment; failures are reported with their diagnosis.

We adopt, and do not claim: low-rank modelling of RIS cascaded channels [19]; the DPSS delay basis [12], [13]; matrix-free iterative equalization through product-convolution operators [17] (here the representation is exact rather than a basis-expansion approximation); learning the quasi-static BS–RIS channel on a slow timescale [20]; Bayesian evidence for model and timing selection [23], [31]; and the observation, known from space-time-coding metasurfaces, that a time shift of the switching pattern multiplies harmonic h by a linear phase [4]. Section II explains how each differs from its use here.

D. Organization and notation

Section II positions the work. Section III gives the system model and every assumption with its justification. Section IV derives the structure of the coupling. Section V presents the receiver, algorithm

by algorithm, with its complexity. Section VI describes the evaluation protocol and baselines, Section VII the results, and Section VIII the trade-offs and limitations.

Bold lower- and upper-case letters denote vectors and matrices; $\circ$ and $\odot$ are the Hadamard product of matrices and vectors; $(\cdot)^\top$, $(\cdot)^\mathsf{H}$ and $(\cdot)^*$ are transpose, conjugate transpose and conjugate; $\mathrm{diag}(\boldsymbol{a})$ is the diagonal matrix with $\boldsymbol{a}$ on its diagonal; $\mathrm{vec}(\cdot)$ stacks columns; $\mathcal{CN}(\boldsymbol{\mu}, \boldsymbol{C})$ is the circularly symmetric complex Gaussian distribution; $\mathrm{sinc}(x) = \sin(\pi x)/(\pi x)$. Confidence intervals are $\pm 1.96\, s/\sqrt{n}$ over n independent channel draws.

II. Related Work and Positioning

A. Time-modulated surfaces and OFDM

Space-time-coding digital metasurfaces [4] control harmonics by the switching pattern: a time delay of an element's pattern multiplies harmonic h by a linear phase, which steers harmonic beams. In communications, time-modulated intelligent reflecting surfaces aligned with OFDM symbols have been designed for waveform security, so that the intended direction receives an undistorted signal and other directions receive scrambled symbols [5]; pattern design has been cast as generative search [6], [9] and extended to sensing [7]. In this line of work the legitimate receiver is a conventional OFDM receiver and the harmonic coupling it sees is either designed to be small in its direction or treated as interference. All of it uses an ideal, frequency-flat element. The only work that undoes the mixing is an attack: an eavesdropper recovers directional-modulation symbols by blind independent component analysis, resolving ambiguities with the Toeplitz structure that the ideal model implies [8]. Rapidly time-varying RIS have also been analysed without OFDM [10]. Iudice *et al.* [1] gave the first physically consistent model with dispersive elements, showed that dispersion breaks the Toeplitz structure, and listed equalization across coupled subcarriers as open.

Closest in spirit outside RIS are OFDM backscatter systems, in which a tag switches its reflection within OFDM symbols: the receiver there decodes the *tag's* data from the switching, or avoids the resulting intercarrier interference with unused subcarriers [25]–[27]; space-time-coding metasurfaces have also been used for sensing from harmonic amplitudes and for non-coherent modulation [28], [29]. Linear periodically time-varying channels are well studied in power-line communication, but with periods much longer than a symbol and hence no harmonic coupling at the subcarrier spacing [30].

Difference. We design the legitimate receiver for the dispersive model: we estimate and equalize the coupling instead of treating it as interference, and our estimator does not rely on a Toeplitz (ideal-element) structure, which dispersion breaks.

B. Estimating non-diagonal OFDM channels

Intra-symbol channel variation from Doppler makes the OFDM channel matrix non-diagonal. The standard estimator expands the time variation on a basis: complex exponentials (CE-BEM) [11], or discrete prolate spheroidal sequences [12], giving a banded frequency-domain matrix with one unknown gain profile per harmonic. Sparse recovery over harmonic–delay dictionaries (orthogonal matching pursuit [14]) is an alternative. These methods assume that the time variation is *unknown*; they must estimate it.

Difference. For a TM-RIS the time variation is known, up to the propagation and the element response. Our estimator uses the known timing and estimates only the static propagation (with the dispersion absorbed). The difference is large in practice: with two pilot symbols CE-BEM reaches 2.2/1.1 bit/s/Hz at 20/30 dB against our 6.5/9.7, and a full-harmonic CE-BEM needs 32 pilot symbols to approach our two-pilot result (Section VII-D). We use CE-BEM, with the same delay prior as ours, as the main baseline.

C. Equalization of non-diagonal OFDM channels

Banded MMSE equalizers keep only the strongest neighbouring subcarriers [15], [16]. Iterative Krylov solvers (LSQR, GMRES) with fast products through product-convolution operators equalize a basis-expansion channel without forming the matrix [17].

Difference. We adopt the iterative approach. For a TM-RIS the product-convolution form is exact (Proposition 1), not a basis-expansion approximation, and for identical patterns all elements share one kernel per state, so a product needs one batched FFT per state. Banded equalizers fail here even with the true channel, because of the $1/h$ harmonic envelope (Section VII-D).

D. RIS channel estimation

Conventional RIS channel estimation assumes a configuration that is constant during each OFDM symbol and changes it between symbols, so that the cascaded channel can be separated [18]; low-rank and sparse structure of the cascaded channel is widely exploited [19]. Two-timescale schemes estimate the quasi-static BS–RIS channel rarely and the mobile RIS–user channel often [20].

Difference. None of these handles switching inside the symbol, where each received subcarrier mixes many transmitted ones. We use low-rank structure and the two-timescale principle, but inside the harmonic-coupled model, where the static side also carries the element dispersion.

E. Synchronization

Timing offsets between RIS reflection patterns and the receiver have been studied for distributed multi-RIS systems with maximum-likelihood estimators [21] and for RIS-assisted symbiotic radio, where the

RIS modulates its own data and an unknown offset causes intercarrier interference that is estimated blindly from virtual subcarriers [22]. Bayesian model selection and model averaging over timing offsets have been used in OFDM channel estimation [23], [24], building on the evidence framework [31].

Difference. We show that for a TM-RIS a clock offset is exactly an opposite delay shift of the two hops. Consequently the pilot residual is nearly flat in the offset, a fact that explains the failure of residual search at high SNR and motivates evidence-based estimation. We use the evidence of our own first-stage model; the tool is known, its necessity and its use here are not.

TABLE I

CAPABILITIES OF EXISTING APPROACHES AND OF THE PROPOSED RECEIVER FOR OFDM THROUGH A TM-RIS. "PILOTS" IS THE NUMBER OF OFDM PILOT SYMBOLS NEEDED TO COME WITHIN 0.3 BIT/S/HZ OF PERFECT CHANNEL KNOWLEDGE AT 20 DB IN OUR SETTING, PRE-REGISTERED FAIR RERUN WITH 30 CHANNELS AND THE ROBUST EQUALIZER FOR EVERY ESTIMATOR (TABLE XI); > 32: NOT REACHED WITH 32.

Approach	in-symbol switching	dispersive element	element model needed	full coupling	pilots	clock offset	cost per $\boldsymbol{Hx}$
One-tap OFDM (TM-IRS designs [5], [6])	coupling as noise	–	–	no	1	–	M
Blind ICA attack [8]	yes	no (Toeplitz)	ideal	partly	0	–	–
Model-aware LMMSE (ours, Sec. VII-B)	yes	yes	exact	yes	2	–	$QM \log M$
Ideal-element LMMSE (Table XI)	yes	no	ideal	yes	> 32	–	$QM \log M$
CE-BEM, $\lvert h\rvert \le 12$ [11]	generic	yes	no	truncated	> 32	–	HM
CE-BEM, all harmonics [11]	generic	yes	no	yes	> 32	–	M^2 (dense), $M \log M$ per shift (CG)
OMP, harmonic×delay [14]	generic	yes	no	truncated	> 32	–	sparse
RIS-OFDM estimation [18]	no	–	–	–	–	–	M
Two-timescale RIS [20]	no	–	–	–	–	–	–
Proposed (timing-only)	**yes**	**yes**	**no**	**yes**	**2 (TDL-C: 4)**	**blind**	$\boldsymbol{QM \log M}$

III. SYSTEM MODEL

A. OFDM through a time-modulated RIS

A base station (BS) transmits OFDM with M active subcarriers at frequencies $f_m = f_c + m\Delta f$, $m \in \mathcal{M} = \{-M/2, \dots, M/2-1\}$, useful duration $T_u = 1/\Delta f$ and cyclic prefix T_{cp}. The signal reaches the user through a RIS of Q elements on a $\sqrt{Q} \times \sqrt{Q}$ grid with half-wavelength spacing. The direct BS–user path is first assumed blocked; Section VII-L removes this assumption.

Element q is a two-state cell controlled by a periodic sequence $s_q[k] \in \{1, 2\}$, $k = 0, \dots, K-1$, with period T_u: it holds state $s_q[k]$ during the slot $[kT_u/K, (k+1)T_u/K)$. The duty cycle of a sequence is

the fraction D of slots in state 1. In state s the element has the single-resonance response [1]

$$\Gamma_s(f) = -1 + \frac{2\xi_{r,s}}{j2\pi(f - f_{0,s}) + \xi_{r,s} + \xi_{i,s}}, \tag{1}$$

with resonance $f_{0,s}$, radiative decay $\xi_{r,s}$ and loss $\xi_{i,s}$. We use the OpenRIS varactor cell fitted in [1, Table I]: $f_{0,1} = 3.471\,\text{GHz}$, $f_{0,2} = 3.710\,\text{GHz}$, $\xi_{r,s}/2\pi = 97.5,\ 128.4\,\text{MHz}$, $\xi_{i,s}/2\pi = 8.9,\ 9.0\,\text{MHz}$. Across a 100 MHz n78 carrier at $f_c = 3.594\,\text{GHz}$ the two states differ by $172°$–$184°$ in phase and by up to 0.57 dB in magnitude, and each state's phase drifts by about $47°$ (Fig. 2). The element is therefore far from the ideal frequency-flat ± 1 switch assumed by earlier TM-RIS work.

B. Input–output relation

Let $a_q^{\mathrm{s}}(f)$ and $a_q^{\mathrm{d}}(f)$ be the frequency responses from the BS to element q and from element q to the user. If the cyclic prefix covers the delay spread of both hops plus the memory of the element response (13.8 ns for this cell), the generalized cyclic-prefix condition of [1] holds, and the received frequency-domain symbol is

$$\boldsymbol{y} = \boldsymbol{H}\boldsymbol{x} + \boldsymbol{w}, \qquad H_{\bar{m}m} = \sum_{q=1}^{Q} a_q^{\mathrm{d}}(f_{\bar{m}})\; b_q^{[\bar{m}-m]}(f_m)\; a_q^{\mathrm{s}}(f_m), \tag{2}$$

with $\boldsymbol{w} \sim \mathcal{CN}(\boldsymbol{0}, \sigma^2 \boldsymbol{I})$ and $b_q^{[h]}(f)$ the h-th Fourier coefficient, over one period, of the switched response of element q evaluated at the input frequency f [1, eq. (5)]. Equation (2) is the ground truth used throughout the evaluation; its implementation was validated against an independent time-domain simulation (filter, gate, FFT) to within 0.27% median and 0.49% maximum relative error (Section VII-A).

C. Propagation models

Shared-path model. Elements of a compact surface see the same propagation paths and differ only by geometric phase. Each hop has one line-of-sight path and $P = 5$ scattered paths with an exponential power-delay profile (30 ns rms) and Rician factor 10 dB; path ℓ with delay τ_ℓ, gain g_ℓ and arrival direction $\boldsymbol{u}_\ell$ gives $a_q(f) = \sum_\ell g_\ell e^{-j\boldsymbol{k}\boldsymbol{u}_\ell^\top \boldsymbol{r}_q} e^{-j2\pi f \tau_\ell}$.

3GPP models. TR 38.901 TDL-A, TDL-C (non-line-of-sight) and TDL-D (line-of-sight) [33], with delay spread 30, 100 or 300 ns; each tap is given a random arrival direction, which sets its phase across the surface, and the line-of-sight tap of TDL-D can use exact spherical distances (near field).

Mobility. When the user moves at speed v, only the RIS–user hop changes. Path ℓ acquires the Doppler shift $\nu_\ell = f_D \cos\phi_\ell$, $f_D = v f_c/c$; it rotates by $e^{j2\pi\nu_\ell n T_s}$ from symbol to symbol ($T_s = T_u + T_{\mathrm{cp}} = 35.67\,\mu\text{s}$) and also mixes subcarriers within each symbol (the standard Doppler intercarrier-interference matrix, included in the ground truth).

D. Assumptions

Table II lists every assumption, why it is reasonable, and where its violation is evaluated. Assumptions A1–A3 define the problem; A4–A10 are relaxed or stress-tested in Section VII.

IV. Structure of the Coupling

This section derives the one structural fact on which the receiver is built and the consequences that each receiver block uses.

Proposition 1 (Timing–propagation separation)**.** *For a piecewise-constant K-slot pattern with period T_u,*

$$b_q^{[h]}(f) = \sum_{s=1}^{2} t_{q,s}[h]\,\Gamma_s(f),$$
$$t_{q,s}[h] = \frac{e^{-j\pi h/K}}{K}\,\mathrm{sinc}\Big(\frac{h}{K}\Big) \sum_{k:\, s_q[k]=s} e^{-j2\pi hk/K}. \tag{3}$$

Hence, with $\boldsymbol{a}_q^{\mathrm{d}} = [a_q^{\mathrm{d}}(f_m)]_m$, $\boldsymbol{a}_q^{\mathrm{s}} = [a_q^{\mathrm{s}}(f_m)]_m$, $\boldsymbol{\Gamma}_s = [\Gamma_s(f_m)]_m$ and Toeplitz matrices $[\boldsymbol{T}_{q,s}]_{\bar{m}m} = t_{q,s}[\bar{m}-m]$,

$$\boldsymbol{H} = \sum_{q=1}^{Q}\sum_{s=1}^{2} \boldsymbol{T}_{q,s} \circ \big(\boldsymbol{a}_q^{\mathrm{d}}(\boldsymbol{\Gamma}_s \odot \boldsymbol{a}_q^{\mathrm{s}})^{\top}\big). \tag{4}$$

Proof. During slot k the element applies $\Gamma_{s_q[k]}(f)$, so its switched response is $\sum_s \Gamma_s(f)\,\mathbb{1}_s(t)$, where $\mathbb{1}_s(t)$ indicates the slots in state s. The h-th Fourier coefficient of the indicator of slot k is

$$\tfrac{1}{K} e^{-j\pi h/K}\,\mathrm{sinc}(h/K)\, e^{-j2\pi hk/K};$$

summing over the slots in state s gives $t_{q,s}[h]$. This is the slot sum of [1, eq. (5)] grouped by state. Substituting into (2) and noting that $t_{q,s}$ depends on $\bar{m}-m$ only gives (4). □

The identity is elementary; its value lies in the following consequences.

Corollary 1 (No element model is needed)**.** *The element response appears in* (4) *only through $\boldsymbol{\Gamma}_s \odot \boldsymbol{a}_q^{\mathrm{s}}$, a per-state* input-side *response that also contains the unknown BS–RIS channel. Estimating $\boldsymbol{v}_{q,s} := \boldsymbol{\Gamma}_s \odot \boldsymbol{a}_q^{\mathrm{s}}$ as an unknown absorbs the dispersion exactly; the receiver needs only $\boldsymbol{T}_{q,s}$, i.e. the switching timing.*

Corollary 2 (Fast products)**.** *$\big(\boldsymbol{T} \circ (\boldsymbol{a}\boldsymbol{b}^{\top})\big)\boldsymbol{x} = \boldsymbol{a} \odot \big(\boldsymbol{T}(\boldsymbol{b} \odot \boldsymbol{x})\big)$, and $\boldsymbol{T}\boldsymbol{z}$ is a linear convolution of length-M sequences. Hence $\boldsymbol{H}\boldsymbol{x}$ and $\boldsymbol{H}^{\mathsf{H}}\boldsymbol{y}$ cost $\mathcal{O}(QM\log M)$ with FFTs of length $\geq 3M-2$, and $\boldsymbol{H}$ never needs to be formed. If all elements use the same pattern, $\boldsymbol{T}_{q,s} = \boldsymbol{T}_s$ for all q, and one batched FFT per state serves all elements.*

TABLE II

ASSUMPTIONS, THEIR JUSTIFICATION, AND WHERE THEIR VIOLATION IS EVALUATED.

	Assumption	Justification	Tested when violated
A1	The receiver knows each element's switching *pattern* $s_q[k]$.	The pattern is chosen by the network (the RIS controller is driven by the BS); for a legitimate user it is control information, like a pilot configuration. Without it, the coupling cannot be told apart from propagation.	Wrong pattern for one of two panels: rate collapses (Sec. VII-L). Stated as a requirement.
A2	Generalized CP: T_{cp} covers both hops' delay spread plus the element memory.	Condition of the anchor model [1]; the cell memory is 13.8 ns, far below the NR normal CP of 2.34 μs.	TDL-D at 300 ns has taps up to 3.8 μs, beyond the CP: out of scope (Sec. VII-I).
A3	Switching period equals T_u and is aligned with the OFDM symbol, *up to an unknown clock offset*.	The anchor model and TM-IRS designs synchronize the pattern with the symbol; any residual misalignment is a clock offset, which we estimate.	Offset estimated blindly (Sec. VII-J); random per-edge jitter evaluated as a hardware requirement.
A4	Two-state elements with responses of the form (1).	Published fit of a real cell [1]. The *estimator does not use* this form (timing-only); it only defines the ground truth.	Element-to-element spread, model mismatch, transients (Sec. VII-B, VII-K).
A5	Elements share propagation paths (compact, far-field surface).	Physical for a surface much smaller than the link distance; it is what makes the unknowns low-rank.	3GPP taps with random directions, up to 1024 elements, near-field user at 3–30 m (Sec. VII-I).
A6	The channel is constant over the pilots and data of one refresh block.	The coherence time is $0.423/f_D$ [32]: 42 ms (1188 symbols) at 3 km/h. The core results isolate switching-induced coupling from Doppler, as in [1].	Users at 3, 30, 120 km/h, refresh interval optimized with pilot overhead (Sec. VII-N).
A7	Common-phase tracking per OFDM symbol (in mobility and phase-noise results).	Standard in 5G NR (phase-tracking reference signals [35]).	Results without any tracking and with an unrealistic per-subcarrier genie are also reported (Sec. VII-N).
A8	Residual CFO, phase noise and switching transients are small or handled.	Standard OFDM synchronization; rise time can be fitted blindly.	CFO, Wiener phase noise, finite rise time (Sec. VII-K).
A9	Noise variance σ^2 known at the receiver.	Standard; it is readily estimated from empty resource elements.	Not varied.
A10	No mutual coupling between elements.	As in the anchor model [1]; requires a full-wave model.	Not tested; stated as a limitation.

Corollary 3 (Harmonic tail)**.** $|t_{q,s}[h]| \le \frac{1}{K}|\mathrm{sinc}(h/K)|\,|\{k : s_q[k] = s\}|$ *decays as* $1/|h|$*, so the coupling energy decays as* $1/h^2$*. For the ideal* ± 1 *element,* $K = 8$ *and* $D = 0.75$*, the fraction of energy outside* $|h| \le H$ *is 9.7%, 5.0%, 3.4%, 1.0% and 0.31% for* $H = 4, 8, 12, 40, 128$ *(numerical evaluation of* (3)*). A model truncated to* $|h| \le H$ *therefore has a normalized error floor of at least* $10\log_{10}$ *of that fraction:* -14.7 *dB at* $H = 12$*, which matches the* -14 *to* -16 *dB floor measured for a truncated estimator (Section VII-D). The tail is a property of ideal (instantaneous) edges: with a finite rise time the coefficients decay faster (0.45% outside* $|h| \le 12$ *for a ramp of 0.3 slot), and truncated estimators recover accordingly (Section VII-D).*

Corollary 4 (Energy on the original subcarrier)**.** *For ideal* ± 1 *elements with a common pattern of duty* D*,* $b^{[0]} = 2D - 1$ *and* $\sum_h |b^{[h]}|^2 = 1$*. A receiver that equalizes subcarrier* m *with the diagonal of* $\boldsymbol{H}$ *alone (one-tap) collects the fraction* $(2D-1)^2$ *of the reflected energy and sees the rest as interference; its signal-to-interference ratio is bounded by* $(2D-1)^2/(1-(2D-1)^2)$*, i.e.* -4.8 *dB at* $D = 0.75$ *and* $+1.1$ *dB at* $D = 0.875$*. Joint equalization collects all of it, an energy-collection gain of* $(2D-1)^{-2}$*: 6.0 dB at* $D = 0.75$ *(Appendix A; measured 5.9 dB with the dispersive cell, Section VII-A).*

Corollary 5 (Shared paths give low rank)**.** *Under A5,* $a_q^{\mathrm{d}}(f) = \sum_{\ell=1}^{P_d} g_\ell e^{-j\boldsymbol{k}\boldsymbol{u}_\ell^\top \boldsymbol{r}_q} e^{-j2\pi f \tau_\ell}$*, so* $[\boldsymbol{a}_1^{\mathrm{d}}, \ldots, \boldsymbol{a}_Q^{\mathrm{d}}]$ *has rank at most* P_d*, the number of user-hop paths, whatever* Q*; the same holds for the input side. With a delay basis* $\boldsymbol{F} \in \mathbb{C}^{M\times L}$ *spanning delays* $[0, \tau_{\max}]$*, the element-domain factors can be written* $\boldsymbol{a}_q^{\mathrm{d}} = \boldsymbol{F}\boldsymbol{U}\boldsymbol{\alpha}_q$ *and* $\boldsymbol{v}_{q,s} = \boldsymbol{F}\boldsymbol{V}_s\boldsymbol{\beta}_q$ *with* $\boldsymbol{U}, \boldsymbol{V}_s \in \mathbb{C}^{L\times r}$ *and* r *on the order of the number of paths.*

Corollary 6 (Groups, shifts and the direct path)**.** *(i) Elements with the same pattern share* $\boldsymbol{T}_{q,s}$*; with* G *distinct patterns,*

$$\boldsymbol{H} = \sum_{g=1}^{G}\sum_{s=1}^{2} \boldsymbol{T}_{g,s} \circ \big((\boldsymbol{F}\boldsymbol{U})\boldsymbol{\Lambda}_g(\boldsymbol{F}\boldsymbol{V}_s)^\top\big), \qquad \boldsymbol{\Lambda}_g = \textstyle\sum_{q\in\mathcal{G}_g} \boldsymbol{\alpha}_q\boldsymbol{\beta}_q^\top \in \mathbb{C}^{r\times r}. \tag{5}$$

(ii) If pattern g *is the base pattern cyclically shifted by* k *slots,* $t_{g,s}[h] = t_{0,s}[h]e^{-j2\pi hk/K}$*, i.e.* $\boldsymbol{T}_{g,s} = \boldsymbol{D}_k\boldsymbol{T}_{0,s}\boldsymbol{D}_k^*$ *with* $\boldsymbol{D}_k = \mathrm{diag}(e^{-j2\pi mk/K})$*; this is the harmonic phase rule of space-time coding [4]. (iii) Since each element is always in one of its two states,* $\boldsymbol{T}_{q,1} + \boldsymbol{T}_{q,2} = \boldsymbol{I}$*; an unswitched (direct) path is the special case of equal input responses in both states, so it is already contained in* (5)*.*

Proposition 2 (A clock offset is an opposite delay of the two hops)**.** *If the switching runs* d *slots late relative to the receiver's timing, every kernel becomes* $\boldsymbol{D}\boldsymbol{T}_{g,s}\boldsymbol{D}^*$ *with* $\boldsymbol{D} = \mathrm{diag}(e^{-j2\pi md/K})$*, and*

$$(\boldsymbol{D}\boldsymbol{T}\boldsymbol{D}^*) \circ (\boldsymbol{a}\boldsymbol{b}^\top) = \boldsymbol{T} \circ \big((\boldsymbol{D}\boldsymbol{a})(\boldsymbol{D}^*\boldsymbol{b})^\top\big). \tag{6}$$

Multiplying a frequency response by $\boldsymbol{D}$ *delays it by* dT_u/K*; by* $\boldsymbol{D}^*$*, advances it.*

Proof. Shifting the pattern by d slots multiplies its h-th coefficient by $e^{-j2\pi hd/K} = e^{-j2\pi \bar{m}d/K} e^{+j2\pi md/K}$ with $h = \bar{m} - m$, which is (6) entrywise. □

Proposition 3 (Physical timing)**.** *Let the BS–RIS and RIS–user hops have bulk delays τ_b and τ_u on all their paths, and let the receiver synchronize its FFT window to the first cascaded arrival. (i) If the RIS controller is locked to the signal arriving at the RIS, the synchronized coupling matrix equals the one without bulk delays: bulk delays have no effect. (ii) If the controller is locked to the BS clock, the synchronized coupling matrix equals the one without bulk delays and with a clock offset $d = -\tau_b/T_{\mathrm{slot}}$.*

Proof. The delay τ_b multiplies the input side by $e^{-j2\pi f_m \tau_b}$ and τ_u the output side by $e^{-j2\pi f_{\bar{m}} \tau_u}$; synchronization multiplies the output by $e^{+j2\pi f_{\bar{m}}(\tau_b+\tau_u)}$. In (i) the switching edges are delayed by τ_b, which by Proposition 2 contributes $e^{-j2\pi(\bar{m}-m)\Delta f \tau_b}$; all phases cancel. In (ii) the edges are not delayed, and the residual phases are those of an offset $-\tau_b$. Both identities hold numerically to $7{\cdot}10^{-15}$ (Section VII-J). □

Remark 1 (When can the pilot residual identify the offset?). By Proposition 2, a receiver that assumes a wrong offset can still fit the pilots with a delayed user hop and an advanced BS hop, *if* its delay basis represents those shifted responses. With the DPSS basis used here ($L = 16$, $\tau_{\max} = 0.3\,\mu$s) the projection error of an advanced response is $1.1 \cdot 10^{-5}$, $2.7 \cdot 10^{-4}$, $1.4 \cdot 10^{-2}$ and 0.82 for shifts of 50, 100, 200 and 500 ns: the residual is nearly flat only for small offsets, which are exactly the physical ones (Proposition 3(ii) gives $|d|T_{\mathrm{slot}} = \tau_b$, typically tens to hundreds of ns). A wrong offset then overfits the pilots: it explains them with improbable responses and predicts held-out pilots poorly. Two criteria resist this overfitting: the pilot residual of a full low-rank estimate evaluated on a fine grid, and the Bayesian evidence of the linear first stage, which penalizes improbable responses explicitly. Section VII-J compares them; the evidence is the cheaper and smoother of the two, not the only one that works.

V. The Receiver

A. Architecture

Fig. 1 shows the complete pipeline. The transmitter sends N_p pilot OFDM symbols followed by data, either as full pilot symbols or as comb pilots with empty neighbouring resource elements (Section VII-M shows the two are equivalent and that data next to pilots is not). The receiver removes the cyclic prefix, applies the FFT and, under mobility, corrects one common phase per symbol (A7). It then (i) builds the timing kernels $\boldsymbol{T}_{g,s}$ from the known pattern; (ii) estimates the RIS clock offset $\hat{d}$ blindly from the pilots (Algorithm 2) and shifts the kernels; (iii) estimates the coupling matrix in factored form (Algorithm 1), either from scratch or warm-started from the previous refresh block (Algorithm 4), optionally with the

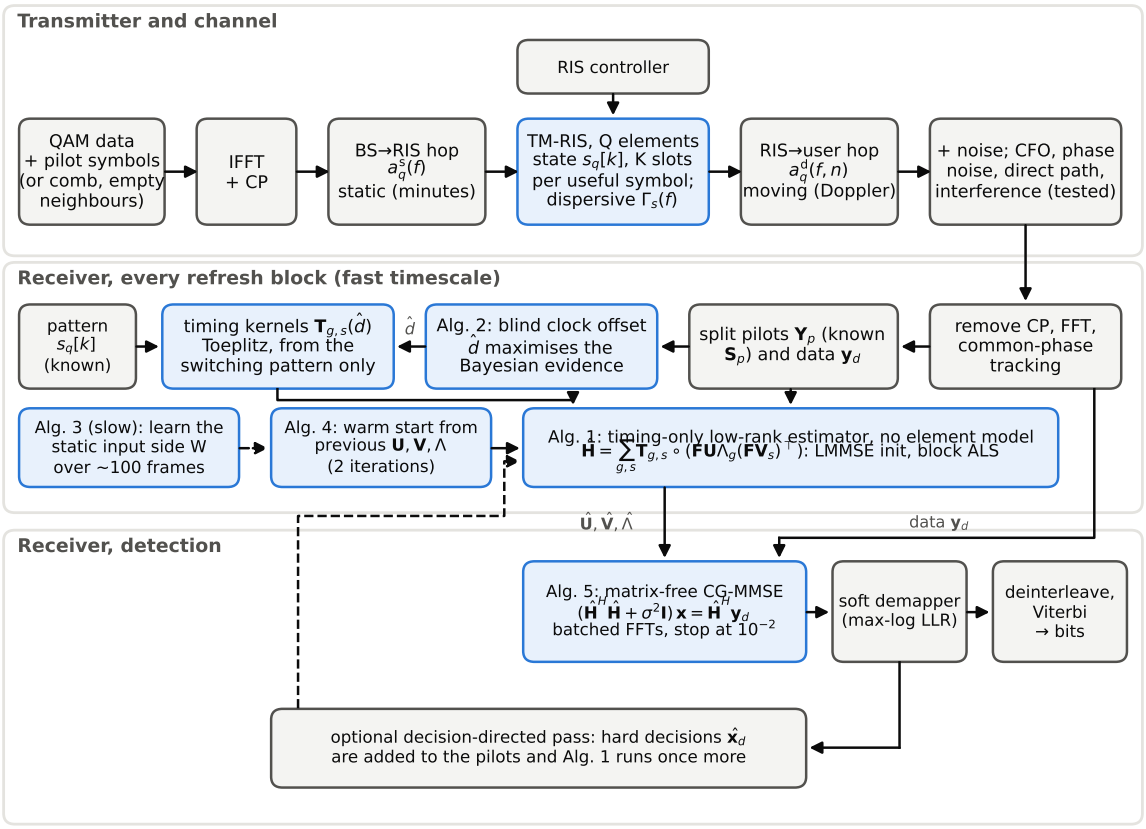


Fig. 1. Complete system and receiver pipeline. Shaded blocks are the proposed receiver components; each is an algorithm of Section V. The receiver uses the switching pattern (A1) and never the element model.

static input side learned on a slow timescale (Algorithm 3); (iv) equalizes the data matrix-free by conjugate gradients (Algorithm 5); and (v) computes max-log LLRs from its own estimate for a soft-input channel decoder. One optional decision-directed (DD) pass reuses hard decisions as additional pilots. All blocks use the factored model (5); the $M \times M$ matrix is never formed.

B. Timing-only low-rank estimation

Model and prior. Each hop's response is delay-limited to $[0, \tau_{\max}]$. We represent it with the first L discrete prolate spheroidal sequences (DPSS) of the bandwidth–delay product $M\Delta f\tau_{\max}$ [12], [13]: $\boldsymbol{F} \in \mathbb{C}^{M\times L}$ holds the leading eigenvectors of $R_{nm} = e^{-j\pi(n-m)\Delta f\tau_{\max}}\mathrm{sinc}((n-m)\Delta f\tau_{\max})$, and $\boldsymbol{\lambda} \in \mathbb{R}^L$ their eigenvalues normalized to unit sum. The DPSS basis is the best L-dimensional subspace for responses with delays uniform in $[0, \tau_{\max}]$ and has no delay grid to fall off. The eigenvalues serve as prior variances: coefficient l of any hop response has prior variance $\varsigma\lambda_l$, with the scale ς set by moment

matching on the pilots. This prior is what keeps the estimator stable when L is larger than the effective number of delay dimensions ($M\Delta f\tau_{\max}+1\approx 3.3$ at $M=256$, $\tau_{\max}=0.3\,\mu$s, while $L=16$).

Observation model. Pilot symbol p is $\boldsymbol{s}_p\in\mathbb{C}^M$ with received $\boldsymbol{y}_p$. By Corollary 2, $\boldsymbol{H}\boldsymbol{s}_p$ is linear in each of $\boldsymbol{U}$, $\boldsymbol{V}_s$ and $\boldsymbol{\Lambda}_g$ when the others are fixed, with regressors built from $\boldsymbol{G}_{g,s,p}=\boldsymbol{T}_{g,s}(\boldsymbol{s}_p\odot\boldsymbol{F})\in\mathbb{C}^{M\times L}$, computed once by FFT:

$$\boldsymbol{H}\boldsymbol{s}_p=\textstyle\sum_g\sum_{a=1}^{r}(\boldsymbol{F}\boldsymbol{u}_a)\odot\big[\sum_s\boldsymbol{G}_{g,s,p}\boldsymbol{V}_s\boldsymbol{\Lambda}_g^{\top}\big]_{:,a} \tag{7}$$

$$=\textstyle\sum_g\sum_s\sum_{b=1}^{r}\big[(\boldsymbol{F}\boldsymbol{U}\boldsymbol{\Lambda}_g)_{:,b}\odot\boldsymbol{G}_{g,s,p}\big]\boldsymbol{v}_{s,b} \tag{8}$$

$$=\textstyle\sum_g\sum_{a,b}\big[(\boldsymbol{F}\boldsymbol{u}_a)\odot(\sum_s\boldsymbol{G}_{g,s,p}\boldsymbol{V}_s)_{:,b}\big]\,[\boldsymbol{\Lambda}_g]_{ab}, \tag{9}$$

which are linear in $\boldsymbol{U}$, in $\boldsymbol{V}_s$ and in $\boldsymbol{\Lambda}_g$, respectively. Stacking the N_p pilots gives, for each block $\boldsymbol{\theta}\in\{\mathrm{vec}\,\boldsymbol{U},\mathrm{vec}[\boldsymbol{V}_1;\boldsymbol{V}_2],\mathrm{vec}\{\boldsymbol{\Lambda}_g\}\}$, a linear model $\boldsymbol{y}=\boldsymbol{A}_{\boldsymbol{\theta}}\boldsymbol{\theta}+\boldsymbol{w}$, $\boldsymbol{y}=\mathrm{vec}[\boldsymbol{y}_1,\dots,\boldsymbol{y}_{N_p}]$. With prior $\boldsymbol{\theta}\sim\mathcal{CN}(\boldsymbol{0},\boldsymbol{P})$, $\boldsymbol{P}=\mathrm{diag}(\boldsymbol{p})$, the LMMSE update is computed in the numerically stable form

$$\hat{\boldsymbol{\theta}}=\boldsymbol{P}^{1/2}\big(\boldsymbol{P}^{1/2}\boldsymbol{A}_{\boldsymbol{\theta}}^{\mathsf{H}}\boldsymbol{A}_{\boldsymbol{\theta}}\boldsymbol{P}^{1/2}+\sigma^2\boldsymbol{I}\big)^{-1}\boldsymbol{P}^{1/2}\boldsymbol{A}_{\boldsymbol{\theta}}^{\mathsf{H}}\boldsymbol{y}, \tag{10}$$

which remains well posed when some prior variances are tiny. The $\boldsymbol{\Lambda}$ step is a convex least-squares problem with a small ridge (10^{-3} of the mean diagonal of $\boldsymbol{A}^{\mathsf{H}}\boldsymbol{A}$).

Initialization. Ignoring the low-rank coupling between groups, $\boldsymbol{H}\approx\sum_s(\sum_g\boldsymbol{T}_{g,s})\circ(\boldsymbol{F}\boldsymbol{C}_s\boldsymbol{F}^{\top})$ is linear in the two $L\times L$ delay matrices $\boldsymbol{C}_s$. Its LMMSE estimate with prior $\varsigma(\boldsymbol{\lambda}\otimes\boldsymbol{\lambda})$ is exact for identical patterns ($G=1$), and its SVD, $[\hat{\boldsymbol{C}}_1\ \hat{\boldsymbol{C}}_2]=\boldsymbol{Y}\boldsymbol{\Sigma}\boldsymbol{W}^{\mathsf{H}}$, initializes $\boldsymbol{U}=\boldsymbol{Y}_{:,1:r}\boldsymbol{\Sigma}_r^{1/2}$ and $[\boldsymbol{V}_1;\boldsymbol{V}_2]=\boldsymbol{W}^{*}_{:,1:r}\boldsymbol{\Sigma}_r^{1/2}$, $\boldsymbol{\Lambda}_g=\boldsymbol{I}$. An earlier version kept the pooled linear estimate as a fallback whenever it fitted the pilots better. The ablation (Section VII-E) showed that this rule is itself an overfitting trap: the linear model has more freedom, fits the pilots better and predicts worse, and on TDL channels at 30 dB it was chosen in 30–57% of channels and increased the gap. On fresh channels (pre-registered), removing it never made the gap worse and reduced it by up to 0.25 ± 0.17 bit/s/Hz; Algorithm 1 therefore always returns the low-rank estimate. The alternating iterations converge within one or two iterations from the SVD initialization, while random initializations are 3–5 times worse, so the initialization matters more than the iteration count.

Identifiability. The number of complex unknowns is $3Lr+Gr^2$, against MN_p observations. Table III gives the counts used in the evaluation and those of the baselines. The nominal counts are close to or above the observations at $M=256$; the estimator is nevertheless well determined because the DPSS prior concentrates each hop on about $M\Delta f\tau_{\max}+1$ effective dimensions. Quantitatively, the effective number of degrees of freedom of the linear first stage, tr of its LMMSE hat matrix, is only 19, 22, 24 and 27 at 20 dB (23, 27, 30, 33 at 30 dB) for 1, 2, 4 and 8 pilot symbols, against 256 to 2048 observations and

Algorithm 1 Timing-only low-rank estimation

Require: pilots $\boldsymbol{Y}, \boldsymbol{S}$; kernels $\{\boldsymbol{T}_{g,s}\}$; DPSS $\boldsymbol{F}, \boldsymbol{\lambda}$; σ^2; rank r; iterations I (20)

$\boldsymbol{G}_{g,s,p} \leftarrow \boldsymbol{T}_{g,s}(\boldsymbol{s}_p \odot \boldsymbol{F})$ by FFT ▷ once

$[\hat{\boldsymbol{C}}_1, \hat{\boldsymbol{C}}_2] \leftarrow$ LMMSE of the pooled linear model, prior $\varsigma(\boldsymbol{\lambda} \otimes \boldsymbol{\lambda})$

$\boldsymbol{U}, \boldsymbol{V}_1, \boldsymbol{V}_2 \leftarrow$ rank-r SVD of $[\hat{\boldsymbol{C}}_1\ \hat{\boldsymbol{C}}_2]$; $\boldsymbol{\Lambda}_g \leftarrow \boldsymbol{I}$; prior of $\boldsymbol{U}, \boldsymbol{V}_s$ columns $\propto \boldsymbol{\lambda}$

for $i = 1, \ldots, I$ **do**

  $\{\boldsymbol{\Lambda}_g\} \leftarrow$ ridge least squares (convex) ▷ (linear in $\boldsymbol{\Lambda}$)

  $\boldsymbol{U} \leftarrow$ LMMSE (10) ▷ Lr unknowns

  $[\boldsymbol{V}_1; \boldsymbol{V}_2] \leftarrow$ LMMSE (10) ▷ $2Lr$ unknowns

end for

$\hat{\boldsymbol{H}} \leftarrow \sum_{g,s} \boldsymbol{T}_{g,s} \circ ((\boldsymbol{F}\boldsymbol{U})\boldsymbol{\Lambda}_g(\boldsymbol{F}\boldsymbol{V}_s)^\top)$ (kept factored)

return $\hat{\boldsymbol{H}}$ ▷ no pilot-fit fallback, see text

512 nominal unknowns: this, not the unknown count, is why one or two pilot symbols suffice. Under the assumed prior, the Bayesian bound on the coupling-matrix error is -32.4 and -39.4 dB with two pilot symbols at 20 and 30 dB; the achieved error is 5–6 dB higher (-26.4, -34.3 dB), because real channels (sparse paths, exponential profiles) do not follow the uniform-delay prior. A learned prior could close part of this gap. At the full band the low-rank form is essential: the linear model has $2L^2 = 1800$ unknowns and its accuracy degrades with bandwidth (Section VII-F), while the low-rank model has $3Lr + r^2 = 784$.

TABLE III

COMPLEX UNKNOWNS OF EACH ESTIMATOR AND PILOT OBSERVATIONS. $L = 16$ AT $M = 256$ (FIXED IN CALIBRATION); $L = \lceil M\Delta f \tau_{\max} \rceil + 10 = 30$ AT THE FULL BAND ($\tau_{\max} = 0.2\,\mu$S); CE-BEM USES A TWO-HOP BASIS WITH $L_2 = \lceil M\Delta f \cdot 0.6\,\mu s \rceil + 12 = 17$.

Estimator	unknowns	observations ($N_p = 2$)
Ours, $M = 256$, $L = 16$, $r = 8$, $G = 1$	$3Lr + Gr^2 = 448$	512
Ours, $M = 256$, $G = 16$ (all patterns distinct)	1408	512
Ours, linear first stage, $M = 256$	$2L^2 = 512$	512
Ours, $M = 3276$, $L = 30$, $r = 8$	784	6552
Ours, linear first stage, $M = 3276$	1800	6552
CE-BEM $\lvert h\rvert \le 12$ / 40 / 128 / all 255	425 / 1377 / 4369 / 8687	512
Unstructured banded LS ($\lvert h\rvert \le 16$)	33 per row	needs $N_p \ge 33$

C. Blind recovery of the RIS clock offset

By Proposition 2 an unknown offset d (in slots) replaces each kernel by $\boldsymbol{D}_d\boldsymbol{T}_{g,s}\boldsymbol{D}_d^*$. For a candidate d, the pooled linear first stage of Algorithm 1 is a linear Gaussian model $\boldsymbol{y} = \boldsymbol{\Phi}_d\boldsymbol{c} + \boldsymbol{w}$, $\boldsymbol{c} \sim \mathcal{CN}(\boldsymbol{0}, \boldsymbol{P})$, $\boldsymbol{P} = \varsigma\,\mathrm{diag}(\boldsymbol{\lambda}\otimes\boldsymbol{\lambda}, \boldsymbol{\lambda}\otimes\boldsymbol{\lambda})$. Its evidence (marginal likelihood) is, up to a constant,

$$\log p(\boldsymbol{y}\,|\,d) = -\boldsymbol{y}^{\mathsf{H}}\boldsymbol{C}_d^{-1}\boldsymbol{y} - \log\det\boldsymbol{C}_d, \qquad \boldsymbol{C}_d = \boldsymbol{\Phi}_d\boldsymbol{P}\boldsymbol{\Phi}_d^{\mathsf{H}} + \sigma^2\boldsymbol{I}. \tag{11}$$

It is evaluated in the parameter domain ($2L^2$ dimensions instead of MN_p) with the Cholesky factor of

$$\boldsymbol{B} = \sigma^{-2}\boldsymbol{P}^{1/2}\boldsymbol{\Phi}_d^{\mathsf{H}}\boldsymbol{\Phi}_d\boldsymbol{P}^{1/2} + \boldsymbol{I} = \boldsymbol{R}^{\mathsf{H}}\boldsymbol{R}:$$

$$\log\det\boldsymbol{C}_d = MN_p\log\sigma^2 + 2\textstyle\sum_i \log R_{ii},$$
$$\boldsymbol{y}^{\mathsf{H}}\boldsymbol{C}_d^{-1}\boldsymbol{y} = \sigma^{-2}\|\boldsymbol{y}\|^2 - \sigma^{-4}\|\boldsymbol{R}^{-\mathsf{H}}\boldsymbol{P}^{1/2}\boldsymbol{\Phi}_d^{\mathsf{H}}\boldsymbol{y}\|^2$$

(Appendix B). The first term rewards fit; the log-determinant penalizes models that can fit too much. A wrong offset fits through improbable shifted responses and loses evidence even when it fits well.

The delay reference. The prior places each hop's delays in $[t_0, t_0 + \tau_{\max}]$. After synchronization the two hops' first arrivals add to zero up to the offset, so a window that starts after the synchronization reference ($t_0 > 0$) cannot hold both hops, and a late synchronization has the same effect. We therefore let the evidence also choose t_0 from a small set, jointly with d (Algorithm 2): the evidence of different priors is directly comparable.

Algorithm 2 Blind clock-offset and delay-reference estimation by evidence

Require: pilots $\boldsymbol{Y}, \boldsymbol{S}$; nominal kernels $\{\boldsymbol{T}_{g,s}\}$; DPSS $\boldsymbol{F}, \boldsymbol{\lambda}$; σ^2; window starts $\mathcal{T}_0 = \{-150, -75, 0, +75\}$ ns

1: **for** each $t_0 \in \mathcal{T}_0$ **do**
2: $\quad\boldsymbol{F}_{t_0} \leftarrow \mathrm{diag}(e^{-j2\pi m\Delta f t_0})\boldsymbol{F}$ ▷ delays $[t_0, t_0 + \tau_{\max}]$
3: $\quad\mathcal{E}_{t_0}(d) \leftarrow$ (11) with kernels $\boldsymbol{D}_d\boldsymbol{T}_{g,s}\boldsymbol{D}_d^*$ and basis $\boldsymbol{F}_{t_0}$
4: $\quad d_0 \leftarrow \arg\max \mathcal{E}_{t_0}(d)$ over $d \in \{-0.5, -0.475, \ldots, 0.5\}$ slot (41 points)
5: $\quad\hat{d}_{t_0} \leftarrow$ golden-section maximization of $\mathcal{E}_{t_0}$ on $[d_0 - 0.025, d_0 + 0.025]$, 14 steps
6: **end for**
7: $\hat{t}_0 \leftarrow \arg\max_{t_0} \mathcal{E}_{t_0}(\hat{d}_{t_0})$; **return** $(\hat{d}_{\hat{t}_0}, \hat{t}_0)$; Algorithm 1 runs with these kernels and basis

D. Two-timescale estimation: learning the static input side

With a fixed BS and RIS, the input-side factors $\boldsymbol{W}_s = \boldsymbol{F}\boldsymbol{V}_s$ (BS–RIS channel and element dispersion) do not change between frames, while $\boldsymbol{U}$ (user hop) does. Learning $\boldsymbol{W}_s$ once from N_f past frames and estimating only $\boldsymbol{U}$ per frame reduces the per-frame unknowns from $3Lr + r^2$ to Lr and makes the per-frame problem linear. The principle is known for conventional RIS [20]; two details are specific to our model. (i) *Initialization from all frames.* Each frame's first-stage estimate is $[\hat{\boldsymbol{C}}_{1,f}\ \hat{\boldsymbol{C}}_{2,f}] = \boldsymbol{U}_f[\boldsymbol{V}_1^\top\ \boldsymbol{V}_2^\top]$, so stacking the frames vertically gives a matrix whose row space is the shared input side; its top-r right singular vectors initialize $\boldsymbol{V}_s$. (ii) *Learned prior.* The learned $\boldsymbol{W}_s$ is an arbitrary invertible mixture of the true input side, so the columns of $\boldsymbol{U}$ are correlated. We learn $\boldsymbol{R}_u = \frac{1}{N_f L}\sum_f \boldsymbol{U}_f^\top \operatorname{diag}(\boldsymbol{\lambda})^{-1}\boldsymbol{U}_f^*$ and use the prior $\operatorname{vec}\boldsymbol{U} \sim \mathcal{CN}(\boldsymbol{0}, \boldsymbol{R}_u \otimes \operatorname{diag}(\boldsymbol{\lambda}))$.

Algorithm 3 Learning the static input side over N_f frames (slow timescale)

Require: pilots of N_f frames $\{\boldsymbol{Y}_f, \boldsymbol{S}_f\}$ sharing the BS–RIS hop; $r = Q$; iterations I_s (6–8)

1: stack $[\hat{\boldsymbol{C}}_{1,f}\ \hat{\boldsymbol{C}}_{2,f}]$ of the first N_{init} frames; $[\boldsymbol{V}_1; \boldsymbol{V}_2] \leftarrow$ top-r right singular vectors
2: **for** $i = 1, \ldots, I_s$ **do**
3: $\quad \boldsymbol{U}_f \leftarrow$ LMMSE given $\boldsymbol{W}_s = \boldsymbol{F}\boldsymbol{V}_s$, prior $\boldsymbol{R}_u \otimes \operatorname{diag}(\boldsymbol{\lambda})$, for every frame
4: $\quad \boldsymbol{R}_u \leftarrow \frac{1}{N_f L}\sum_f \boldsymbol{U}_f^\top \operatorname{diag}(\boldsymbol{\lambda})^{-1}\boldsymbol{U}_f^*$
5: $\quad [\boldsymbol{V}_1; \boldsymbol{V}_2] \leftarrow$ LMMSE over all frames jointly (normal equations accumulated frame by frame)
6: **end for**
7: **return** $\boldsymbol{W}_s = \boldsymbol{F}\boldsymbol{V}_s$, $\boldsymbol{R}_u$. Per new frame: $\boldsymbol{U} \leftarrow$ LMMSE given $\boldsymbol{W}_s$ (one linear solve)

E. Warm-started updates

When the channel is refreshed every N symbols for a moving user, consecutive estimates are close. Algorithm 4 starts from the previous factors and runs two ALS iterations, skipping the first stage, which dominates the cost of Algorithm 1. A cheaper variant keeps $\boldsymbol{V}_s$ fixed and updates only $\boldsymbol{\Lambda}$ and $\boldsymbol{U}$; its trade-off is quantified in Section VII-P.

Algorithm 4 Warm-started refresh

Require: new pilots; previous $\boldsymbol{U}, \boldsymbol{V}_s, \boldsymbol{\Lambda}_g$; iterations I_w (2); flag *freezeV*

1: **for** $i = 1, \ldots, I_w$ **do**
2: $\quad \{\boldsymbol{\Lambda}_g\} \leftarrow$ ridge LS; $\boldsymbol{U} \leftarrow$ LMMSE; **if not** *freezeV*: $[\boldsymbol{V}_1; \boldsymbol{V}_2] \leftarrow$ LMMSE
3: **end for**

F. Matrix-free equalization

The MMSE estimate of the data symbol solves $(\hat{\boldsymbol{H}}^{\mathsf{H}}\hat{\boldsymbol{H}}+\sigma^2\boldsymbol{I})\boldsymbol{x}=\hat{\boldsymbol{H}}^{\mathsf{H}}\boldsymbol{y}_d$. We use conjugate gradients with products through (5) (Corollary 2), as in [17]. For $G=1$ the product is $\hat{\boldsymbol{H}}\boldsymbol{x}=\sum_a(\boldsymbol{F}\boldsymbol{u}'_a)\odot\mathcal{F}^{-1}\big[\sum_s\mathcal{F}(\boldsymbol{t}_s)\odot\mathcal{F}((\boldsymbol{F}\boldsymbol{v}_{s,a})\odot\boldsymbol{x})\big]$, where $\boldsymbol{u}'_a$ are the columns of $\boldsymbol{U}\boldsymbol{\Lambda}$: the forward FFTs of all r columns (and all data symbols) are one batched call per state, the two states are summed in the frequency domain, and one batched inverse FFT follows.

Algorithm 5 Matrix-free CG-MMSE equalization

Require: $\boldsymbol{y}_d$ (one or several symbols as columns); factored $\hat{\boldsymbol{H}}$; σ^2; tolerance $\epsilon=10^{-2}$; $i_{\max}=50$

$\boldsymbol{b}\leftarrow\hat{\boldsymbol{H}}^{\mathsf{H}}\boldsymbol{y}_d$; $\boldsymbol{x}\leftarrow\boldsymbol{0}$; $\boldsymbol{r}\leftarrow\boldsymbol{b}$; $\boldsymbol{p}\leftarrow\boldsymbol{r}$

for $i=1,\dots,i_{\max}$ **do**

  $\boldsymbol{q}\leftarrow\hat{\boldsymbol{H}}^{\mathsf{H}}(\hat{\boldsymbol{H}}\boldsymbol{p})+\sigma^2\boldsymbol{p}$ ▷ two batched FFT products

  $\alpha\leftarrow\|\boldsymbol{r}\|^2/\boldsymbol{p}^{\mathsf{H}}\boldsymbol{q}$; $\boldsymbol{x}\leftarrow\boldsymbol{x}+\alpha\boldsymbol{p}$; $\boldsymbol{r}_{\mathrm{new}}\leftarrow\boldsymbol{r}-\alpha\boldsymbol{q}$

  if $\|\boldsymbol{r}_{\mathrm{new}}\|/\|\boldsymbol{b}\|<\epsilon$ for every column **then break**

  $\boldsymbol{p}\leftarrow\boldsymbol{r}_{\mathrm{new}}+(\|\boldsymbol{r}_{\mathrm{new}}\|^2/\|\boldsymbol{r}\|^2)\boldsymbol{p}$; $\boldsymbol{r}\leftarrow\boldsymbol{r}_{\mathrm{new}}$

end for

return $\boldsymbol{x}$ (columnwise scalars in practice)

G. Complexity

Per ALS iteration of Algorithm 1, forming and solving the normal equations of the three blocks costs (complex multiply–adds)

$$C_{\mathrm{it}}=\sum_{n\in\{Lr,\,2Lr,\,Gr^2\}}\Big(MN_pn^2+\frac{n^3}{3}\Big), \tag{12}$$

and the regressors cost $2GN_pL(N_{\mathrm{f}}\log_2 N_{\mathrm{f}}+2N_{\mathrm{f}})$ once (N_{f} the FFT length). The first stage adds $MN_p(2L^2)^2+(2L^2)^3/3$. One evidence evaluation in Algorithm 2 costs about the same as the first stage, and the search uses 55 of them per window start (220 for four starts). One CG iteration costs two products of $\mathcal{O}(rM\log M)$ each. Table IV evaluates these expressions; Section VII-P gives measured times.

VI. Evaluation Methodology

A. Setup

Table V lists the default parameters. Results at the full 100 MHz n78 carrier use $M=3276$; most comparisons use $M=256$ with the same subcarrier spacing, which preserves the per-subcarrier coupling

TABLE IV

OPERATIONS (COMPLEX MULTIPLY–ADDS) PER ESTIMATE, FROM (12) WITH $r = 8$, $N_p = 2$, 20 ITERATIONS, $\tau_{\max} = 0.2\,\mu$S, AND OF THE ALTERNATIVES.

	$M = 256$	$M = 1024$	$M = 3276$
Algorithm 1, $G = 1$	$5.7 \cdot 10^8$	$4.1 \cdot 10^9$	$3.9 \cdot 10^{10}$
Algorithm 1, $G = 16$	$1.8 \cdot 10^{10}$	$5.4 \cdot 10^{10}$	$1.8 \cdot 10^{11}$
Dense MMSE solve (one symbol block)	–	–	$4.7 \cdot 10^{10}$
CG equalizer, per symbol (25 it.)	–	–	$4.2 \cdot 10^8$
Dense $\boldsymbol{H}$ storage	1 MB	17 MB	172 MB

structure at a cost that permits 10–30 independent channels per point. Only the central half of the band is scored, so that band-edge effects do not favour any method. The receiver never uses the ground-truth channel or the element model, except in the explicitly labelled genie and oracle references.

TABLE V

DEFAULT EVALUATION PARAMETERS.

Carrier, spacing	$f_c = 3.594$ GHz (n78), $\Delta f = 30$ kHz, $T_{\mathrm{cp}} = 2.34\,\mu$s
Subcarriers	$M = 256$ (central 128 scored); full band $M = 3276$ (central 1638)
Surface	$Q = 16$ elements (4×4, $\lambda/2$); up to 1024 tested
Switching	$K = 8$ slots per T_u, duty $D = 0.75$, identical patterns (S1) unless stated
Element	OpenRIS cell, two states, [1, Table I]
Channel	shared paths, 1 LoS + 5 NLoS per hop, 30 ns rms, $K_{\mathrm{R}} = 10$ dB; 3GPP TDL-A/C/D
Pilots	$N_p = 2$ QPSK pilot symbols unless stated
Estimator	DPSS, $\tau_{\max} = 0.3\,\mu$s, $L = 16$ ($M = 256$); $\tau_{\max} = 0.2\,\mu$s, $L = 30$ ($M = 3276$); $r = 8$; 20 iterations
TDL setting	$\tau_{\max} = \min(T_{\mathrm{cp}}, 10\times$ delay spread), $L = \lceil M\Delta f \tau_{\max} \rceil + 10$, $r = 16$
Equalizer	MMSE (direct at $M = 256$, CG at $M = 3276$)
SNR	$\mathrm{SNR} = P_c/\sigma^2$, P_c the mean received power per subcarrier

B. Metrics

Achievable rate. For an estimate $\hat{\boldsymbol{H}}$, the MMSE filter $\boldsymbol{W} = (\hat{\boldsymbol{H}}^{\mathsf{H}}\hat{\boldsymbol{H}} + \sigma^2\boldsymbol{I})^{-1}\hat{\boldsymbol{H}}^{\mathsf{H}}$ is applied to the *true* channel; with $\boldsymbol{G} = \boldsymbol{W}\boldsymbol{H}$, subcarrier m has

$$\mathrm{SINR}_m = \frac{|G_{mm}|^2}{\sum_{n \neq m} |G_{mn}|^2 + \sigma^2 \|\boldsymbol{w}_m\|^2}, \qquad R = \frac{1}{|\mathcal{M}_c|} \sum_{m \in \mathcal{M}_c} \log_2(1 + \mathrm{SINR}_m), \tag{13}$$

over the scored set $\mathcal{M}_c$. The *gap* is $R(\boldsymbol{H}) - R(\hat{\boldsymbol{H}})$, with $R(\boldsymbol{H})$ the rate with perfect knowledge. Where losses are quoted in dB, they are the rate gap divided by the local slope of R versus SNR (0.31 and 0.33 bit/s/Hz per dB at 20 and 30 dB). Under mobility, (13) treats G_{mm} as known to the detector, which amounts to per-subcarrier gain tracking; we therefore also report two other versions. With *common-phase tracking* the receiver corrects one complex gain $a = \hat{\boldsymbol{g}}^{\mathsf{H}}\boldsymbol{g}/\hat{\boldsymbol{g}}^{\mathsf{H}}\hat{\boldsymbol{g}}$ per symbol, where $\boldsymbol{g} = \mathrm{diag}(\boldsymbol{G})$ and $\hat{\boldsymbol{g}} = \mathrm{diag}(\boldsymbol{W}\hat{\boldsymbol{H}})$, and the residual $|g_m - a\hat{g}_m|^2$ adds to the interference. With *no tracking* it trusts $\hat{\boldsymbol{g}}$ directly. Mobility results use common-phase tracking as the main metric (A7).

Error rates. Uncoded symbol error rate (SER) with QAM; coded block error rate (BLER) with a rate-1/2, constraint-length-7 convolutional code (generators 133, 171 octal), a random interleaver, Gray-mapped 16-QAM, max-log LLRs computed from each receiver's *own* estimate, and soft Viterbi decoding, 506 information bits per block.

Net throughput. With pilots on the first N_p of every N symbols, the net rate is $R_{\mathrm{net}} = \frac{N-N_p}{N} \cdot \frac{1}{N-N_p}\sum_{n=N_p}^{N-1} R(n)$, where $R(n)$ is the rate at symbol n of the block (the estimate ages with n); N and N_p are optimized per case.

C. Baselines

All estimators use the same pilots, noise and channels, and all model-based baselines get the same LMMSE/DPSS prior as ours, so that differences come from the model, not the regularization.

- **One-tap**: per-subcarrier least squares from the pilots and one-tap equalization: today's OFDM receiver.
- **Toeplitz mixing**: the ideal-element model $H_{\bar{m}m} = t(\bar{m}-m)\,c(m)$ implied by frequency-flat switching, as assumed in [8], estimated with our prior.
- **Ideal-element LMMSE**: the strongest ideal-element baseline, $\boldsymbol{H} = (\boldsymbol{T}_1 - \boldsymbol{T}_2) \circ (\boldsymbol{F}\boldsymbol{C}\boldsymbol{F}^{\mathsf{T}})$ with the ideal ± 1 element and our two-sided DPSS prior (Table XI).
- **CE-BEM** [11]: $\boldsymbol{H} = \sum_{|h|\le H_b} \boldsymbol{S}_h\,\mathrm{diag}(\boldsymbol{F}_2\boldsymbol{c}_h)$ with $\boldsymbol{S}_h$ the shift by h and a two-hop DPSS basis ($0.6\,\mu$s, $L_2 = 17$); $H_b = 4, 8, 12, 40, 128$ and all 255 harmonics; N_p up to 32.
- **OMP** [14] over a harmonic$\times$delay dictionary, stopping at the noise level or 300 atoms.
- **Unstructured banded LS**: every entry within $|h| \le 16$ estimated independently, with 32 orthogonal pilot symbols (16 times our overhead).
- **Model-aware LMMSE** (a variant of ours): the coupling built from the *exact* or a mismatched element model, to measure the value of not needing one.
- **Equalizer fairness**: in the pre-registered rerun (Table XI) every estimator is given the naive MMSE, a robust MMSE with the error power from the pilot residual, and a robust MMSE with the oracle

error power.

- **Equalizers** fed the same estimate and the true channel (oracle): banded MMSE with window $\pm W$ [15], [16]; LSQR/CG [17]; dense MMSE.
- **References**: perfect channel knowledge (upper bound); genie channel without phase-noise knowledge (Section VII-K).

Deep-learning estimators were not included: they require training data matched to the switching pattern and element, and the question studied here is the value of model structure.

D. Pilots only versus decision-directed passes

Results are *pilots only* unless stated. The exceptions are: the element-knowledge comparison (Table VII), the delay-spread check and the uncoded BER experiment, which used two pilot symbols plus *two* decision-directed iterations on six data symbols (16-QAM at 20 dB, 64-QAM at 30 dB); and the full-band results (Table XVI), which report pilots only and one DD pass side by side. The recommendation of at most one DD pass comes from the full band with 64-QAM, where a second pass increased the error rate. Pilots-only versions of the two-iteration experiments have not yet been run and are listed in the claims ledger.

E. Protocol

Each point averages 6–30 independent channel draws (stated per experiment), with 95% confidence intervals. Calibration (DPSS window and regularization) was done once on a separate seed and frozen. For each experiment added after the core results, a pass/fail criterion was written down *before* running it; failed criteria are reported as failures, with diagnosis, and follow-up methods were given new pre-registered criteria. Appendix D maps every table and figure to its script and result file.

VII. Results

A. Model validation

Element model. Our implementation of (1) with the published parameters reproduces the dispersion reported in [1] (Fig. 2): differential phase $172°$–$184°$ across 98 MHz (reported $180° \pm 6.5°$), magnitude imbalance ± 0.57 dB (reported < 0.75 dB), per-state phase variation $46.5°/49.2°$ (reported $\approx 47°/50°$), ripple 0.62/0.50 dB (reported ≈ 0.6 dB).

Coupling model. For $K = 2$ the coupling follows the $\mathrm{sinc}(h/2)$ envelope with even harmonics exactly zero for an ideal element; the dispersive cell creates a DC harmonic at -19 to -27 dB relative to $h = 1$ (Fig. 3). The closed form (2) agrees with an independent time-domain simulation (per-element filtering,

sampled gating, FFT; $Q = 4$, $K = 4$ random patterns, multipath on both hops) to 0.27% median and 0.49% maximum relative error, the residual being the sampling of the gates.

Energy collection. Corollary 4 predicts the gain of joint equalization over the one-tap receiver as $(2D-1)^{-2}$. Measured with the dispersive cell (Q=16, K=8, 30 channels, Fig. 4): 11.6, 5.9 and 2.4 dB at $D = 0.625$, 0.75 and 0.875 against predicted 12.0, 6.0 and 2.5 dB. At 20 dB with perfect knowledge, the one-tap rate is 0.10, 0.42 and 1.19 bit/s/Hz at these duty cycles, the joint MMSE rate 6.14, 6.19 and 6.11, and the matched-filter bound 6.45, 6.52 and 6.43. Joint MMSE recovers about 95% of the matched-filter bound. With random per-element shifts the matrix is worse conditioned and joint MMSE reaches 5.13 against a bound of 6.62 bit/s/Hz at $D = 0.75$.

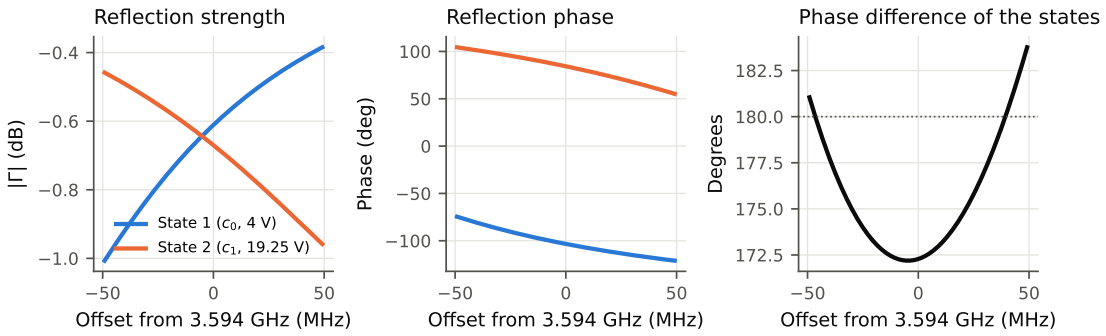


Fig. 2. Reflection magnitude, phase, and phase difference of the two states of the OpenRIS cell across the n78 carrier.

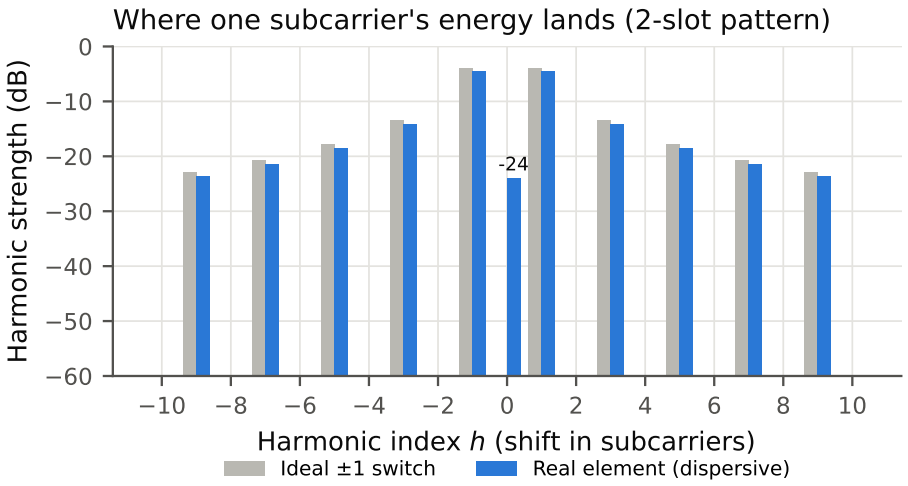


Fig. 3. Harmonic magnitudes for an ideal switch and the dispersive cell ($K = 2$).

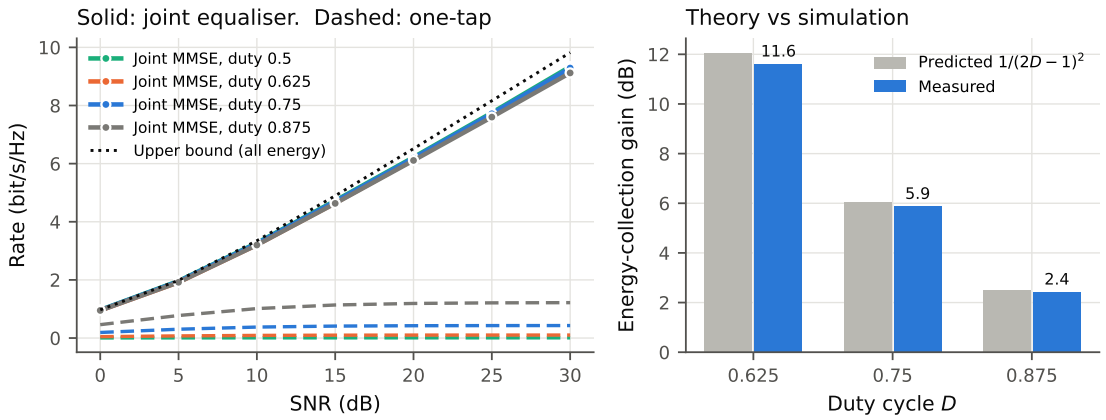


Fig. 4. Left: rate of joint MMSE and one-tap versus SNR for four duty cycles. Right: predicted $(2D-1)^{-2}$ versus measured energy-collection gain.

B. Value of not needing the element model

Table VII compares receivers that differ only in what they assume about the element (identical patterns, two pilot symbols plus two DD iterations, loss in dB against perfect knowledge; fresh seed, 30 channels, 95% CI on the rate loss $\leq \pm 0.05$ bit/s/Hz). A model-aware estimator with the exact model is excellent, but its accuracy collapses at high SNR as soon as the model is wrong: a 10 MHz resonance error costs 2.4 dB at 30 dB, and the frequency-flat ± 1 model of earlier TM-RIS work costs 6.6 dB. The timing-only estimator, which uses no element model at all, is within 0.04–0.21 dB at every SNR (Fig. 5). This is the practical core of Corollary 1: the receiver needs only the switching timing.

Could the receiver fit the element model instead? A natural alternative is to assume the single-pole family and fit its six parameters (resonance and two decay rates per state) from the pilots. Starting from parameters off by $\mathcal{N}(0, 10\,\mathrm{MHz})$ and $\mathcal{N}(0, 10\%)$, we maximized the evidence of the model-aware LMMSE over the six parameters (30 channels, Table VI). With the correct family, the fitted model is no better than the timing-only estimator ($+0.012 \pm 0.022$ and $+0.079 \pm 0.047$ bit/s/Hz worse at 20/30 dB); when the elements are not identical (10 MHz spread), the fit fails ($+0.73 \pm 0.43$ at 30 dB) while the timing-only estimator is unaffected. The model-aware estimators use the linear (full delay-matrix) form; part of the timing-only estimator's advantage at 30 dB comes from its low-rank step, which the model-aware form could also use.

TABLE VI

TIMING-ONLY VERSUS FITTING THE ELEMENT MODEL (GAP TO PERFECT CSI, BIT/S/HZ, 20 / 30 DB, 30 CHANNELS, $N_p = 2$).

Truth	timing-only	oracle model	wrong model	fitted model
Exact single-pole cell	**0.075 / 0.193**	0.083 / 0.269	0.551 / 2.077	0.087 / 0.272
10 MHz element spread	**0.079 / 0.217**	0.915 / 2.700	1.107 / 3.159	0.246 / 0.951

TABLE VII

LOSS (DB) AGAINST PERFECT CHANNEL KNOWLEDGE FOR DIFFERENT ELEMENT KNOWLEDGE. FIRST RUN / CONFIRMATION RUN ON A FRESH SEED (30 CHANNELS).

Receiver assumes	10 dB	20 dB	30 dB
Exact element model	0.02	0.05 / 0.06	0.12 / 0.16
Resonance off by 5 MHz	–	0.53	2.59
Resonance off by 10 MHz	0.06	0.66 / 0.47	3.36 / 2.44
Resonance off by 20 MHz	–	0.55	2.91
Ideal ± 1 switch (earlier TM-RIS work)	0.18	1.41 / 1.42	6.57 / 6.60
No model, timing only (proposed)	**0.04**	**0.08 / 0.10**	**0.11 / 0.21**

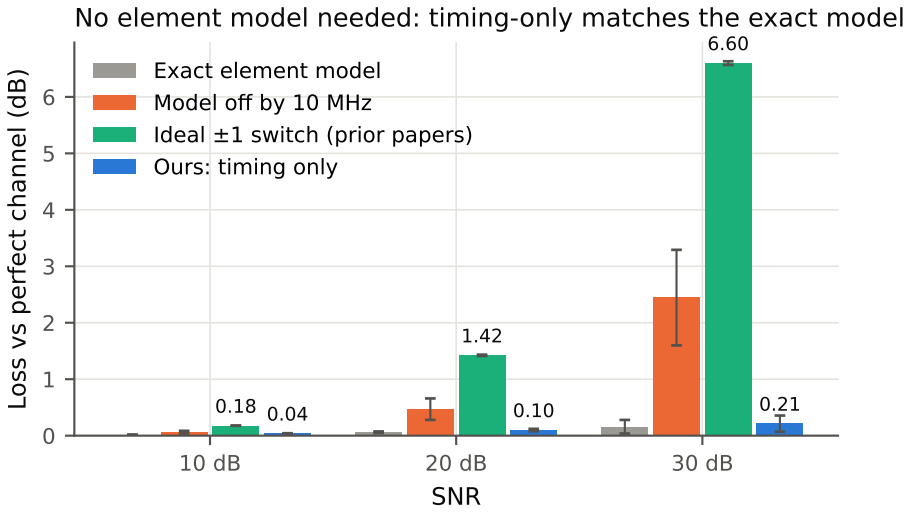


Fig. 5. Loss against perfect knowledge for four assumptions about the element.

C. Is the result an artefact of the model? Independent truth and model mismatch

The ground truth (2) and the estimator's model share the same algebra (Proposition 1), so a reader may suspect an "inverse crime". We therefore ran the estimator *unchanged* on data that do not come from (2) or that deliberately violate its structure (30 channels each, paired against the same channel without the change; pre-registered criterion: loss within 0.3 bit/s/Hz).

Independent ground truth. (i) Received pilots from a sampled-gating time-domain simulation (per-element filtering at the absolute frequency, gating per slot, FFT; $Q = 16$, $K = 8$, multipath on both hops), scored on the matrix obtained by probing the simulator; it differs from (2) by $5.6 \cdot 10^{-3}$ (relative Frobenius norm). (ii) The element's switching transients, which (2) (a quasi-static model) omits: for a single-pole cell the internal state is continuous at each switch and decays with the new state's pole, which adds $H_{\mathrm{tr}}(\bar{m}, m) = \sum_q a_q^{\mathrm{d}}(f_{\bar{m}}) \sum_{\text{edges } k} \frac{2\xi_{r,\mathrm{new}}}{T_u} G_{\mathrm{new}}(f_{\bar{m}}) e^{-j2\pi(\bar{m}-m)\Delta f t_k} [G_{\mathrm{old}}(f_m) - G_{\mathrm{new}}(f_m)] a_q^{\mathrm{s}}(f_m)$ with $G_s(f) = (j2\pi(f - f_{0,s}) + \xi_s)^{-1}$; its relative size is $1.3 \cdot 10^{-3}$. The losses are $+0.006 \pm 0.009$ (time domain) and $+0.003 \pm 0.006$ bit/s/Hz (transients) at 30 dB.

Physics outside the model (Table VIII). A state-dependent output response, mutual coupling that depends on the states of *both* neighbouring elements, frequency-dependent array phases (beam squint) and per-element resonance spread change $\boldsymbol{H}$ by up to 40%, yet the loss stays within $+0.05$ bit/s/Hz everywhere. We had predicted that joint-state coupling would break the model for per-element patterns; it does not at these levels, because the per-group factors absorb it. State-independent coupling is absorbed exactly by construction (it only redistributes the output responses).

TABLE VIII

ESTIMATOR UNCHANGED UNDER MODEL MISMATCH: RELATIVE CHANGE OF $\boldsymbol{H}$ AND LOSS VS THE MATCHED CHANNEL (BIT/S/HZ, PAIRED, T-BASED 95% CI; 30 CHANNELS).

Mismatch	$\|\Delta\boldsymbol{H}\|/\|\boldsymbol{H}\|$	20 dB	30 dB
Time-domain simulation as truth	0.006	-0.004 ± 0.005	$+0.006 \pm 0.009$
Element switching transients	0.001	$+0.001 \pm 0.001$	$+0.003 \pm 0.006$
State-dependent output, $\epsilon = 0.05/0.1/0.2$	0.10/0.17/0.40	$\leq +0.004$	$+0.015/+0.039/+0.034$
Joint-state coupling, $G = 1$, $\kappa = 0.02/0.05/0.08$	0.09/0.28/0.29	$\leq +0.002$	$+0.014/+0.008/+0.015$
Joint-state coupling, $G = 4$, $\kappa = 0.02/0.05/0.08$	0.04/0.13/0.25	$\leq +0.001$	$+0.009/+0.003/-0.025$
Beam squint, $Q = 16$ / $Q = 256$	0.007/0.016	$\leq +0.001$	$0.000/-0.002$
Resonance spread 10 MHz per element	0.18	$+0.003 \pm 0.008$	$+0.046 \pm 0.044$

D. Estimators and baselines

Against the literature's estimators. Table IX and Fig. 6 compare estimators on the same pilots and channels (20 channels, $M = 256$). The one-tap receiver is capped at its interference ratio (Corollary 4). The Toeplitz model of the ideal element in its usual form $H_{\bar{m}m} = t(\bar{m}-m)c(m)$ reaches 2.3/1.4 bit/s/Hz: dispersion breaks its structure (a much stronger ideal-element baseline, with our two-sided delay prior, is in Table XI). CE-BEM with up to ± 12 harmonics reaches at most 2.3/1.4 and barely improves with pilots: by Corollary 3 it misses 3.4% of the coupling energy at ± 12. It even degrades from 20 to 30 dB, because an MMSE filter that trusts a wrong model hurts more when noise no longer masks the model error. The proposed estimator is within 0.09 bit/s/Hz (20 dB) and 0.24 bit/s/Hz (30 dB) of perfect knowledge with two pilot symbols.

TABLE IX

RATE (BIT/S/HZ) WITH EACH ESTIMATOR, 20 CHANNELS, 95% CI. N_p = PILOT SYMBOLS.

	20 dB		30 dB	
Estimator	$N_p = 2$	$N_p = 8$	$N_p = 2$	$N_p = 8$
One-tap	0.42	0.42	0.43	0.43
Toeplitz (ideal element)	2.27 ± 0.08	2.46 ± 0.10	1.42 ± 0.10	1.52 ± 0.03
CE-BEM $\lvert h\rvert \leq 4$	1.29 ± 0.04	1.33 ± 0.03	0.57 ± 0.05	0.58 ± 0.02
CE-BEM $\lvert h\rvert \leq 8$	1.82 ± 0.06	1.88 ± 0.04	0.81 ± 0.07	0.99 ± 0.04
CE-BEM $\lvert h\rvert \leq 12$	2.20 ± 0.08	2.32 ± 0.04	1.08 ± 0.10	1.38 ± 0.06
Proposed	$\mathbf{6.52 \pm 0.02}$	$\mathbf{6.58 \pm 0.02}$	$\mathbf{9.68 \pm 0.08}$	$\mathbf{9.87 \pm 0.02}$
Perfect knowledge	6.61	6.61	9.92	9.92

Strongest baselines. A reviewer may object that CE-BEM failed only because it was truncated. Table X gives CE-BEM with up to all 255 harmonics and up to 32 pilot symbols (20 channels; all harmonics on 4 channels because of its 8687 unknowns). Full-harmonic CE-BEM does approach our result, but only with 32 pilot symbols (6.18/9.37 bit/s/Hz): the proposed estimator with two pilot symbols (6.51/9.69) is better than full CE-BEM with 32, which still falls 0.41/0.52 bit/s/Hz short of perfect knowledge. The correct statement is therefore that ours with two pilot symbols exceeds full CE-BEM with 16 times more pilots; CE-BEM did not reach within 0.3 bit/s/Hz of perfect knowledge at any pilot count tested. At equal overhead ($N_p = 2$) full CE-BEM collapses to 0.13/0.08, because its 8687 unknowns cannot be determined from 512 observations. Sparse recovery (OMP) behaves like truncated CE-BEM (2.32/1.19), and unstructured banded least squares is worst even with 32 orthogonal pilot symbols (0.85/0.54).

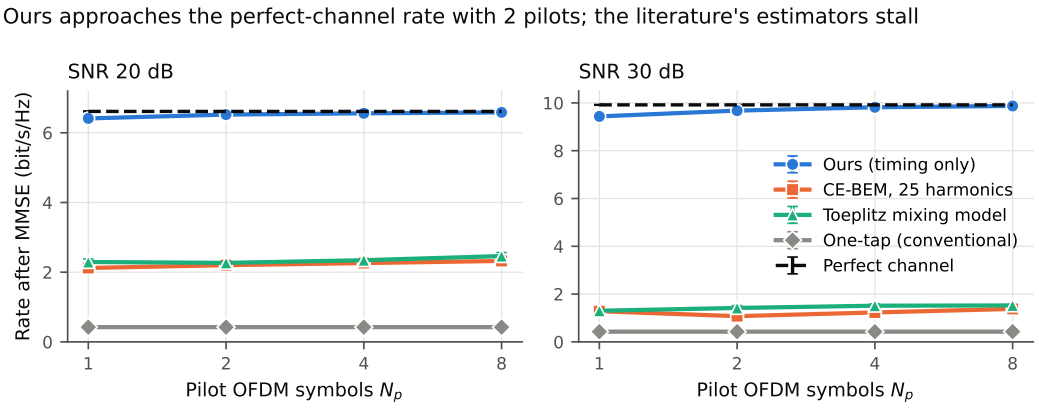


Fig. 6. Rate versus pilot symbols at 20 and 30 dB (20 channels, 95% CI).

TABLE X

STRONGEST BASELINES: RATE (BIT/S/HZ) AT 20 / 30 DB.

	$N_p = 2$	$N_p = 8$	$N_p = 32$
CE-BEM $\lvert h\rvert \le 12$	2.13 / 1.09	2.26 / 1.26	2.33 / 1.41
CE-BEM $\lvert h\rvert \le 40$	2.06 / 1.82	4.20 / 3.34	4.44 / 3.82
CE-BEM $\lvert h\rvert \le 128$	0.15 / 0.07	4.25 / 6.06	6.08 / 8.15
CE-BEM all 255 harmonics	0.13 / 0.08	2.97 / 4.19	6.18 / 9.37
OMP (harmonic×delay)	2.32 / 1.19	–	–
Unstructured banded LS	–	–	0.85 / 0.54
Proposed	**6.51 / 9.69**	–	–
Perfect knowledge	6.59 / 9.89		

Pre-registered fair rerun. Tables IX and X give every baseline the naive MMSE equalizer. Table XI repeats the comparison under rules fixed before the run: 30 channels per point, both the shared-path channel and 3GPP TDL-C at 100 ns, $N_p \in \{2, 4, 8, 16, 32\}$, CE-BEM with no harmonic cap (matrix-free LMMSE, checked against the dense solver to $7 \cdot 10^{-9}$), and three equalizers for *every* estimator: naive MMSE, robust MMSE with the estimation-error power taken from the pilot residual (usable in practice), and robust MMSE with the oracle error power. The Toeplitz baseline is replaced by the strongest ideal-element estimator we could build: the model-aware LMMSE with the ideal ± 1 element, $\boldsymbol{H} = (\boldsymbol{T}_1 - \boldsymbol{T}_2) \circ (\boldsymbol{F}\boldsymbol{C}\boldsymbol{F}^{\mathsf{T}})$, the same prior as ours. Table XI reports the loss to perfect knowledge with the residual-based robust equalizer; the oracle equalizer changes no conclusion (it helps truncated CE-BEM

and OMP by up to 1.5 bit/s/Hz at 30 dB, and ours by at most 0.12).

The ranking survives. Ours is the only estimator that comes within 0.3 bit/s/Hz of perfect knowledge at 20 dB with at most 32 pilot symbols: it needs $N_p = 2$ on the shared-path channel and $N_p = 4$ on TDL-C; no baseline gets there with 32 (Table I). The best baseline at every point is the ideal-element LMMSE: it loses only 0.4–0.5 bit/s/Hz at 20 dB, so the ideal element is a fair approximation when noise dominates, but 2.0–2.3 at 30 dB, where the dispersion it ignores becomes the dominant error, and more pilots do not help it (a model error, not an estimation error). Against the best baseline *per channel* (minimum over the six baselines and three equalizers, paired over channels), ours gains 0.41 ± 0.02 / 2.16 ± 0.04 bit/s/Hz at 20/30 dB with $N_p = 2$ on the shared-path channel and 0.30 ± 0.06 / 1.14 ± 0.25 on TDL-C, and 0.39–0.43 / 0.54–2.15 at all other pilot counts; every interval excludes zero. Uncapped CE-BEM reaches the ideal-element LMMSE only with 32 pilot symbols (loss 0.43 at 20 dB on the shared-path channel), so "ours with two pilot symbols is better than full CE-BEM with 32" also holds under the fair rules. With the estimated-gain detector (the receiver does not know G_{mm}) every loss grows by at most 0.15 bit/s/Hz and the ranking is unchanged.

One pre-registered sanity check failed: the oracle-robust rate of the *truncated* CE-BEM sets ($|h| \le 12$, $|h| \le 40$) still decreases from 20 to 30 dB on the shared-path channel, by up to 0.28 bit/s/Hz. A scalar error power cannot represent a truncation error that is concentrated on the missing harmonics, so scalar robustification does not fully remove the effect described above; the uncapped sets behave as expected (their rate increases with SNR), and the check passes on TDL-C.

Equalizers. Table XII feeds the same estimate, and the true channel, to equalizers from the high-mobility OFDM literature. Banded MMSE fails even with the true channel: a ± 32 band given the true channel reaches 6.21 against 9.89 bit/s/Hz for the dense MMSE at 30 dB. This is not only the harmonic tail: with smooth switching edges, whose tail is much shorter, the banded equalizers do not recover (Table XIII). At $D = 0.75$ three quarters of the energy is off the diagonal, and even a few percent of it outside the window caps the SINR at high SNR. The iterative matrix-free solver matches the dense MMSE in 10 iterations, which justifies Algorithm 5.

Dependence on the switching edges. Table XIII repeats the comparison with a linear switching ramp in the truth (and in our calibrated kernels). CE-BEM's failure depends strongly on sharp edges: with a ramp of 0.3 slot (1.25 μs) CE-BEM with $|h| \le 40$ and 8 pilot symbols reaches 8.2 bit/s/Hz, close to ours with 2 (8.56). For realistic rise times up to 0.05 slot (about 200 ns) our advantage stays above 4 bit/s/Hz at 30 dB. The advantage over truncated estimators is therefore a property of fast switching, which the emission analysis in Section VII-K argues should be slowed to about 0.015 slot or more; we report it as such.

TABLE XI

PRE-REGISTERED FAIR COMPARISON: LOSS TO PERFECT KNOWLEDGE (BIT/S/HZ, MEAN $\pm$ 95% t-INTERVAL, 30 CHANNELS), ROBUST MMSE WITH THE RESIDUAL-BASED ERROR POWER FOR EVERY ESTIMATOR, KNOWN-G_{mm} DETECTOR. PERFECT KNOWLEDGE: 6.62/9.92 (SHARED PATH), 6.20/9.38 (TDL-C). "–": FEWER OBSERVATIONS THAN UNKNOWNS, NOT RUN.

	20 dB			30 dB		
Estimator	$N_p = 2$	$N_p = 8$	$N_p = 32$	$N_p = 2$	$N_p = 8$	$N_p = 32$
Shared-path channel, 30 ns						
One-tap	6.19 ± 0.02	6.19 ± 0.02	6.19 ± 0.02	9.49 ± 0.02	9.49 ± 0.02	9.49 ± 0.02
Ideal-element LMMSE	0.48 ± 0.01	0.44 ± 0.00	0.43 ± 0.00	2.27 ± 0.03	2.16 ± 0.01	2.14 ± 0.01
CE-BEM $\lvert h\rvert \leq 12$	3.77 ± 0.09	3.18 ± 0.05	3.03 ± 0.03	7.26 ± 0.10	6.73 ± 0.06	6.52 ± 0.05
CE-BEM $\lvert h\rvert \leq 40$	4.46 ± 0.13	2.15 ± 0.05	1.84 ± 0.03	8.10 ± 0.18	5.89 ± 0.11	5.35 ± 0.04
CE-BEM $\lvert h\rvert \leq 128$	6.55 ± 0.04	2.34 ± 0.16	0.46 ± 0.01	9.89 ± 0.03	3.60 ± 0.19	1.51 ± 0.04
CE-BEM, all harmonics	–	3.42 ± 0.21	0.43 ± 0.02	–	5.10 ± 0.31	0.55 ± 0.04
OMP	4.16 ± 0.10	3.13 ± 0.05	2.92 ± 0.02	7.49 ± 0.12	6.67 ± 0.06	6.44 ± 0.03
Proposed	$\mathbf{0.07 \pm 0.02}$	$\mathbf{0.01 \pm 0.00}$	$\mathbf{0.00 \pm 0.00}$	$\mathbf{0.11 \pm 0.03}$	$\mathbf{0.02 \pm 0.00}$	$\mathbf{0.01 \pm 0.00}$
3GPP TDL-C, 100 ns						
One-tap	5.79 ± 0.11	5.79 ± 0.11	5.79 ± 0.11	8.97 ± 0.16	8.97 ± 0.16	8.97 ± 0.16
Ideal-element LMMSE	0.85 ± 0.12	0.50 ± 0.02	0.41 ± 0.01	3.05 ± 0.24	2.12 ± 0.08	1.95 ± 0.07
CE-BEM $\lvert h\rvert \leq 12$	4.43 ± 0.13	3.16 ± 0.10	2.81 ± 0.09	7.85 ± 0.18	6.39 ± 0.17	6.02 ± 0.18
CE-BEM $\lvert h\rvert \leq 40$	5.83 ± 0.09	2.71 ± 0.15	1.71 ± 0.12	9.19 ± 0.15	6.03 ± 0.28	4.68 ± 0.26
CE-BEM $\lvert h\rvert \leq 128$	6.05 ± 0.10	5.31 ± 0.09	1.02 ± 0.04	9.30 ± 0.16	8.83 ± 0.13	2.14 ± 0.18
CE-BEM, all harmonics	–	5.58 ± 0.09	1.17 ± 0.03	–	9.04 ± 0.14	1.55 ± 0.09
OMP	4.13 ± 0.14	2.90 ± 0.11	2.59 ± 0.10	7.61 ± 0.20	6.12 ± 0.19	5.80 ± 0.20
Proposed	$\mathbf{0.52 \pm 0.11}$	$\mathbf{0.08 \pm 0.01}$	$\mathbf{0.02 \pm 0.00}$	$\mathbf{1.80 \pm 0.39}$	$\mathbf{0.18 \pm 0.03}$	$\mathbf{0.03 \pm 0.00}$

Fair-case check. Earlier TM-IRS designs choose the pattern so that the intended user sees little coupling. Table XIV uses per-element polarities that co-phase the $h = 0$ harmonic at the user. Even then the one-tap receiver saturates at its interference ratio (3.47 bit/s/Hz at 30 dB for $D = 0.875$, interference ratio $+10.0$ dB), while the proposed receiver continues to 9.37. At 10 dB the gain is modest (+0.6 bit/s/Hz, about 2 dB).

TABLE XII

EQUALIZERS ON THE PROPOSED ESTIMATE (TRUE CHANNEL IN BRACKETS), RATE (BIT/S/HZ).

Equalizer	20 dB	30 dB
One-tap	0.42 (0.42)	0.43 (0.43)
Banded MMSE ± 4	3.23 (3.24)	3.35 (3.35)
Banded MMSE ± 16	4.82 (4.83)	5.26 (5.26)
Banded MMSE ± 32	5.45 (5.47)	6.20 (6.21)
LSQR/CG, 10 iterations	6.51 (6.59)	9.66 (9.83)
Dense MMSE	6.51 (6.59)	9.69 (9.89)

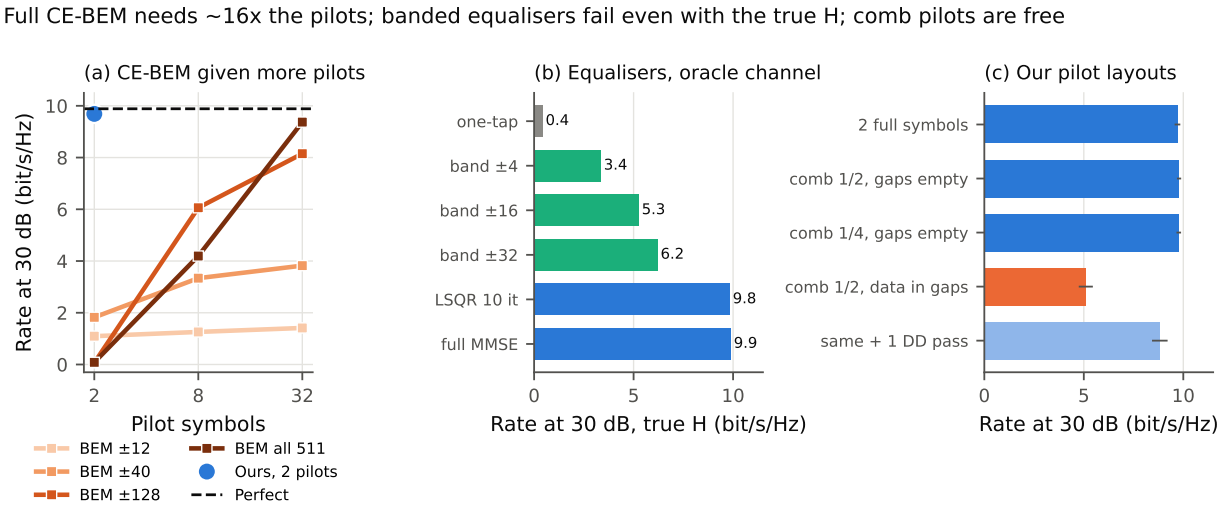


Fig. 7. (a) CE-BEM with more harmonics and pilots; (b) equalizers with the true channel; (c) pilot layouts.

TABLE XIII

RATE AT 30 DB (BIT/S/HZ) VERSUS THE SWITCHING RISE TIME (FRACTION OF A SLOT), 30 CHANNELS.

Rise time	0	0.01	0.05	0.1	0.3
Energy outside $\lvert h\rvert \le 12$	3.4%	3.2%	2.6%	2.0%	0.45%
Banded MMSE ± 32, true channel	6.21	6.20	6.08	5.80	5.20
CE-BEM $\lvert h\rvert \le 12$, $N_p = 8$, robust	3.51	3.43	3.54	3.84	6.32
CE-BEM $\lvert h\rvert \le 40$, $N_p = 8$, robust	4.66	4.60	5.50	7.52	8.19
Proposed, $N_p = 2$	9.69	9.80	9.62	9.17	8.56
Perfect CSI	9.85	9.90	9.80	9.45	8.72

TABLE XIV

USER THE PATTERN IS DESIGNED FOR: ONE-TAP VERSUS JOINT MMSE WITH THE *model-aware* RANK-1 ESTIMATOR (EXACT ELEMENT MODEL, $N_p = 2$), BIT/S/HZ AT 10/20/30 DB. THIS EXPERIMENT PREDATES THE TIMING-ONLY ESTIMATOR AND HAS NOT YET BEEN RERUN WITH IT.

Duty, user	one-tap SIR	one-tap	proposed
0.875, designed	$+10.0$ dB	2.51 / 3.34 / 3.47	3.13 / 6.18 / 9.37
0.75, designed	$+3.5$ dB	1.44 / 1.70 / 1.74	2.77 / 5.63 / 8.80
0.625, designed	-2.3 dB	0.62 / 0.70 / 0.71	2.24 / 4.53 / 7.38
0.75, generic	-12.3 dB	0.17 / 0.19 / 0.19	2.24 / 4.77 / 7.66

E. What each part of the receiver contributes

Table XV ablates the estimator (30 channels; shared-path 30 ns and TDL-C 100 ns). The timing-only linear first stage alone already does most of the work: its gap is 0.10/0.23 bit/s/Hz at 20/30 dB with two pilot symbols on the shared-path channel, where CE-BEM is about 2 bit/s/Hz below perfect knowledge. The low-rank step adds 0.04–0.12 there and 0.15–0.73 on TDL-C, where the linear model has more effective unknowns. The rank matters little between 4 and 16 (on the shared-path channel $r = 2$ is marginally best). One decision-directed pass removes a further 50–70% of the gap at 20 dB. The pilot-fit fallback of an earlier version was removed (Section V-B).

TABLE XV

ABLATION: GAP TO PERFECT CSI (BIT/S/HZ) AT 20 / 30 DB. FULL = LOW RANK WITH THE (SINCE REMOVED) PILOT-FIT FALLBACK.

Channel	N_p	linear only	low rank $r = 2$	$r = 8$	$r = 16$	+ 1 DD pass
Shared-path 30 ns	2	0.104 / 0.229	0.046 / 0.104	0.065 / 0.108	0.090 / 0.124	0.019 / 0.029
	4	0.050 / 0.108	0.026 / 0.054	0.030 / 0.049	0.036 / 0.053	0.014 / 0.019
	8	0.026 / 0.051	0.014 / 0.027	0.015 / 0.024	0.017 / 0.025	0.009 / 0.014
TDL-C 100 ns	2	0.494 / 2.086	0.329 / 1.810	0.342 / 1.359	0.397 / 1.433	0.131 / 1.159
	4	0.248 / 0.966	0.171 / 0.857	0.149 / 0.589	0.174 / 0.583	0.076 / 0.385
	8	0.125 / 0.265	0.088 / 0.432	0.069 / 0.166	0.075 / 0.141	0.048 / 0.108

F. Full band

At $M = 3276$ all estimation and equalization use only the FFT operator. With the linear first stage alone ($2L^2 = 1800$ unknowns) the estimate is noise-limited at -23 to -29 dB at 28 dB SNR (it reaches -45 to -49 dB at 60 dB SNR, so the model is right); the low-rank form reaches -39 dB with two pilot symbols. Table XVI gives SER over 30 independent channels, with 95% intervals of the SER *ratio* to perfect knowledge from a clustered bootstrap (channels resampled, 10^4 resamples). With pilots only, the proposed receiver's SER is 1.10 [1.06, 1.15] times that of perfect knowledge at 20 dB/16-QAM and 1.37 [1.18, 1.86] times at 28 dB/64-QAM; one DD pass brings these to 1.05 [1.03, 1.08] and 1.12 [1.08, 1.21]. (An earlier 10-channel run gave 1.54 at 28 dB pilots-only; the 30-channel interval supersedes it.) Both numbers are reported because the DD figure alone would overstate the pilots-only performance. A second DD pass made 64-QAM worse in an earlier run (decision errors feed back), so we use at most one. With per-element cyclic shifts, the shift structure (Corollary 6) gives an operator error of -34.4 dB at 28 dB and SER ratios of 1.10 and 1.12 (5 channels).

TABLE XVI

FULL BAND ($M = 3276$), 30 CHANNELS: MEAN SER AND, IN BRACKETS, THE SER RATIO TO PERFECT KNOWLEDGE WITH ITS 95% CLUSTERED-BOOTSTRAP INTERVAL.

Receiver	20 dB, 16-QAM	28 dB, 64-QAM
One-tap	0.843	0.956
Perfect knowledge (CG-MMSE)	0.031	0.027
Proposed, pilots only	0.034 (1.10 [1.06, 1.15])	0.037 (1.37 [1.18, 1.86])
Proposed, + 1 DD pass	0.033 (1.05 [1.03, 1.08])	0.031 (1.12 [1.08, 1.21])
CE-BEM ($\lvert h\rvert \le 8$), pilots only	0.568 (18 [13, 29])	0.884 (32 [20, 73])

G. Coded performance

Table XVII gives the block error rate with the convolutional code, measured with 200 channels $\times$ 10 blocks = 2000 blocks per SNR point near the waterfall. The one-tap receiver never decodes a block. CE-BEM with $|h| \le 12$ floors at 0.34 and never reaches 10% up to 14 dB; with $|h| \le 40$ it fails completely with two pilot symbols (too many unknowns), with or without the robust equalizer. The proposed receiver crosses 10% BLER at 8.51 dB [8.37, 8.70] against 8.37 dB [8.23, 8.55] with perfect knowledge, a gap of **0.14 dB [0.09, 0.20]** (log-linear interpolation, 95% clustered-bootstrap intervals over channels). An earlier 120-block estimate (0.4 dB) is superseded.

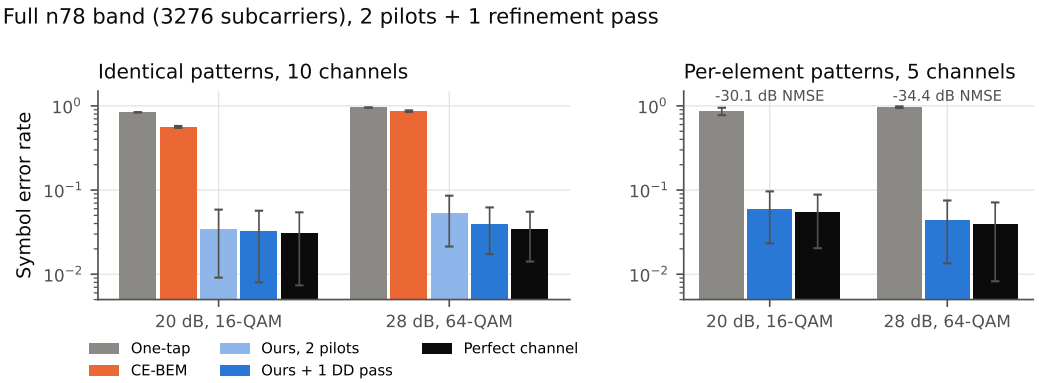


Fig. 8. Full-band symbol error rate for identical and per-element patterns.

TABLE XVII

CODED BLOCK ERROR RATE (RATE-1/2 CONVOLUTIONAL CODE, 16-QAM, $N_p = 2$), 2000 BLOCKS PER POINT.

SNR (dB)	6	7	8	9	10	11	12	14
One-tap	1	1	1	1	1	1	1	1
CE-BEM $\lvert h\rvert \le 12$	1	1	0.985	0.891	0.713	0.531	0.381	0.338
CE-BEM $\lvert h\rvert \le 40$ (also robust eq.)	1	1	1	1	1	1	0.997	0.985
Proposed	0.957	0.625	0.210	0.050	0.018	0.006	0.003	0
Perfect knowledge	0.926	0.559	0.173	0.040	0.013	0.005	0.003	0

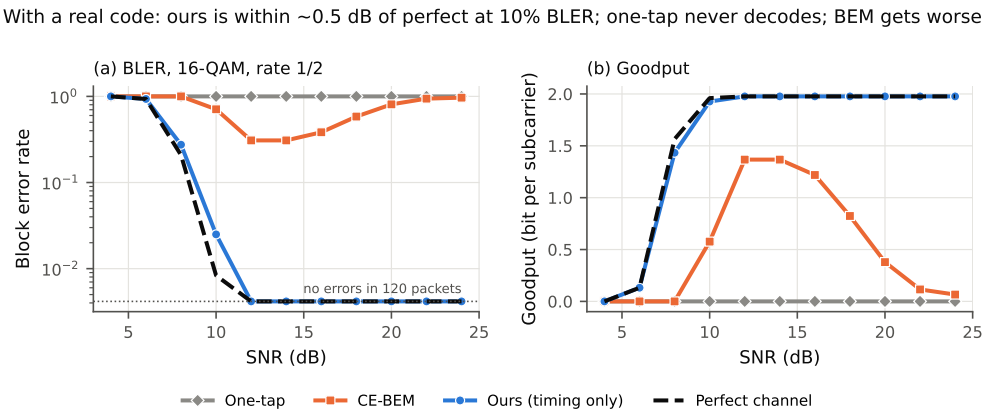


Fig. 9. Coded block error rate and goodput (earlier 120-block run; Table XVII gives the 2000-block measurement).

H. Sensitivity to design parameters

Every parameter we chose was varied one at a time around the default (12 channels per point, Fig. 10). The gap to perfect knowledge stays at 0.06–0.09 / 0.10–0.20 bit/s/Hz (20/30 dB) for $Q = 4$ to 144 elements, 0.06–0.09 / 0.11–0.16 for $K = 2$ to 16 slots, and 0.05–0.13 / 0.12–0.23 for duty cycles 1/8 to 7/8. The one parameter that matters is the number G of *distinct* patterns: with two pilot symbols the gap grows from 0.07 / 0.10 at $G = 1$ to 0.32 / 1.03 at $G = 16$, because the unknowns grow with Gr^2 (Table III); four pilot symbols bring it to 0.19 / 0.38. This gives a design rule: use four pilot symbols when the surface uses many distinct per-element patterns, as security-oriented designs do.

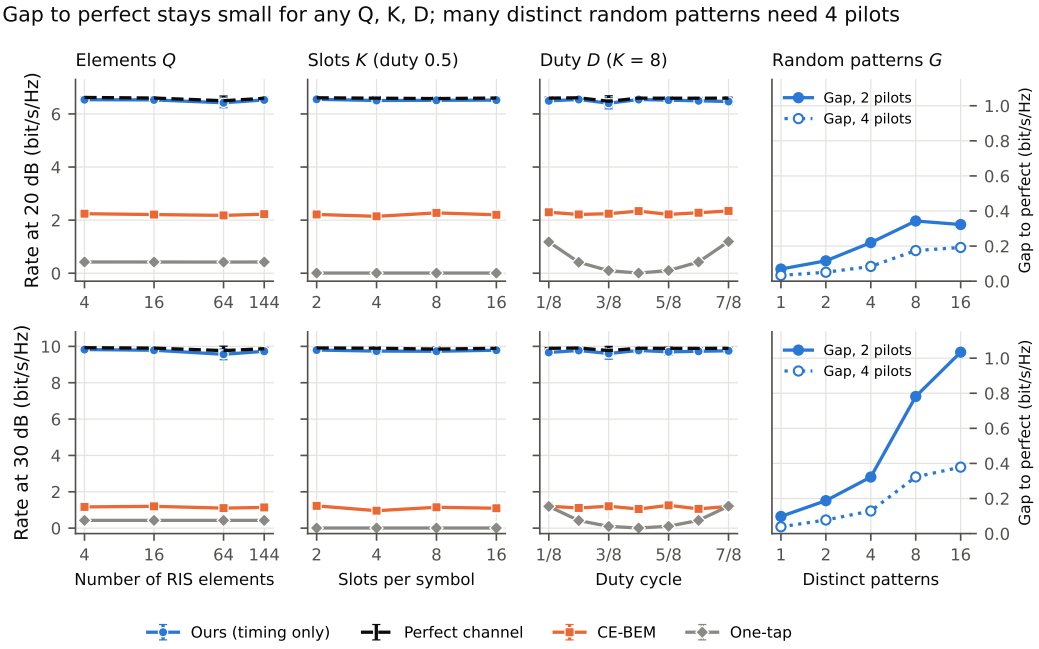


Fig. 10. Sensitivity to the number of elements, slots, duty cycle and distinct patterns.

I. Channel models, surface size and long delay spread

3GPP channels. Table XVIII uses the TR 38.901 TDL models. At 30 ns, as in the anchor paper, two pilot symbols suffice (gap ≤ 0.14 at 20 dB). With realistic delay spreads the default setting, designed for the six-path model, breaks: TDL channels have 14–24 taps from different directions. Matching the delay window to the delay spread ($\tau_{\max} = \min(T_{\mathrm{cp}}, 10{\times}\mathrm{DS})$) and raising the rank to 16 fixes 100 ns with eight pilot symbols: gap 0.08–0.24 bit/s/Hz at 20 dB. The rank barely matters (8, 16 and 32 give the same within 0.1); the number of pilots is the lever, because a longer delay window means more

unknowns (Table III). We note a correction this exposed: an earlier robustness check to 100 ns, made with the six-path model, was optimistic for standard channels.

TABLE XVIII

GAP (BIT/S/HZ) ON 3GPP TDL CHANNELS AT 20 DB (30 DB IN BRACKETS), WINDOW MATCHED TO THE DELAY SPREAD, RANGES OVER RANKS 8/16/32, 6 CHANNELS.

	$N_p = 2$	$N_p = 4$	$N_p = 8$
TDL-A, 100 ns	0.69–0.76 (2.5–2.8)	0.21–0.32 (0.8–0.9)	**0.08–0.09** (0.24–0.27)
TDL-C, 100 ns	0.47–0.50 (1.7–1.9)	0.17–0.24 (0.6–0.8)	**0.09–0.10** (0.18–0.26)
TDL-D, 100 ns	0.65–0.69 (2.1–2.3)	0.31–0.37 (1.3–1.4)	**0.23–0.24** (0.85–0.95)
TDL-A/C/D, 300 ns	1.1–1.5 (3.4–4.7)	0.6–1.1 (2.6–3.3)	0.35–0.92 (1.4–2.3)

Spatially consistent CDL channels. Replacing the TDL taps with random directions by the 38.901 CDL-A/C/D models (cluster angles, 20 rays per cluster) changes little: with eight pilot symbols at 20 dB the gap is 0.087, 0.084 and 0.240 on CDL-A/C/D against 0.085, 0.088 and 0.217 on the TDL versions (30 channels).

How low is the rank? Corollary 5 bounds the rank of the element-domain matrix by the number of paths, but 3GPP channels have hundreds of rays and wideband arrays have beam squint. Table XIX gives the measured effective rank (99% of the energy) of the true RIS–user matrix for CDL channels: it grows from about 5 to 16 when the surface grows from 16 to 1024 elements over 98 MHz with squint, and stays near 5–8 in narrow bands and in the near field. The low-rank property is therefore strong for large surfaces; for $Q = 16$ (rank about 5 of 16) it is weak, and the estimator's gain there comes mostly from the timing-only structure (Table XV).

TABLE XIX

EFFECTIVE RANK (99% ENERGY) OF THE TRUE ELEMENT-DOMAIN MATRIX, CDL 100 NS, 10 DRAWS.

Q	7.7 MHz	98 MHz	98 MHz, squint	near field 3 m	near field 10 m
16	3.5	5.0	5.0	3.2	3.5
64	4.9	8.9	8.8	4.9	4.6
256	5.3	11.9	12.9	6.1	6.6
1024	5.5	14.2	16.0	7.5	7.4

Surface size and near field. With TDL-C at 30 ns, $Q = 64$, 256 and 1024 elements give a gap of

0.06–0.11 / 0.17–0.25 bit/s/Hz at 20/30 dB with two pilot symbols: the unknowns do not grow with Q (Corollary 5). With $Q = 1024$ (1.3 m aperture, Fraunhofer distance about 40 m) and the TDL-D line-of-sight path computed from exact element distances, a user at 3, 10 and 30 m sees a gap of 0.09–0.18 / 0.38–0.60 (Fig. 11).

Cyclic-prefix validity. The prefix condition (A2) applies to the *cascade* of the two hops, whose delays add. Table XX gives, for each TDL profile and delay spread, the maximum cascaded delay and the fraction of cascaded energy that falls beyond the prefix (2.34 μs, plus the 13.8 ns element memory). All 30 ns and 100 ns cases are valid (TDL-D at 100 ns has a single weak tap pair beyond the prefix, at -56 dB). All 300 ns cases are *invalid*: 0.3–1.9% of the energy (-25 to -17 dB) arrives after the prefix, and the ground truth (2) omits the resulting inter-symbol interference. The 300 ns results below are therefore kept only as a record and are not claims of this paper; a valid long-delay study needs an extended prefix or a realistic split of the delay spread between the hops.

TABLE XX

CASCADED DELAY OF THE TWO HOPS AND ENERGY BEYOND THE CYCLIC PREFIX (BOTH HOPS WITH THE SAME TDL PROFILE AND DELAY SPREAD).

Profile, delay spread	max cascaded delay	energy beyond CP	valid
TDL-A/C/D, 30 ns	0.58 / 0.52 / 0.75 μs	none	yes
TDL-A/C, 100 ns	1.93 / 1.73 μs	none	yes
TDL-D, 100 ns	2.51 μs	-56 dB	yes
TDL-A/C/D, 300 ns	5.80 / 5.19 / 7.51 μs	-25.3 / -23.1 / -17.2 dB	no

300 ns (invalid, see above): learning the static side. At 300 ns, eight pilot symbols are not enough. With Algorithm 3 (input side learned from 100 frames of other users or earlier positions, $r = Q = 16$), TDL-A and TDL-C reach a gap of 0.23 and 0.26 bit/s/Hz at 20 dB, against 0.36 and 0.46 from one frame, and the gap at 30 dB halves (Table XXI, Fig. 12). The pre-registered criterion (all three models ≤ 0.3 at 20 dB) nevertheless *failed*, on TDL-D: it fails even when the true static side is given (0.56), because at 300 ns TDL-D has taps up to $12.5 \times 300\,\text{ns} = 3.8\,\mu$s, beyond the 2.34 μs cyclic prefix (A2). Taps beyond the prefix carry about -20 dB of power, which no prefix-limited delay window can represent; a real system would switch to an extended prefix. Two earlier versions of the learning step also failed their criteria: initializing from a single frame gave no gain over one frame (0.37 vs 0.33 on TDL-A), and 30 frames gave 0.31. The stacked initialization, the learned prior and 100 frames were each needed.

One hundred frames is a short time compared with the coherence of the BS–RIS hop, which is static for minutes.

TABLE XXI

300 NS DELAY SPREAD, EIGHT PILOT SYMBOLS: GAP (BIT/S/HZ) AT 20 / 30 DB, 6 CHANNELS.

	one frame	learned side, 100 frames	true static side
TDL-A	0.36 / 1.91	**0.23** / 0.85	0.12 / 0.37
TDL-C	0.46 / 1.84	**0.26** / 0.99	0.12 / 0.32
TDL-D	0.85 / 2.74	0.64 / 2.89	0.56 / 1.61

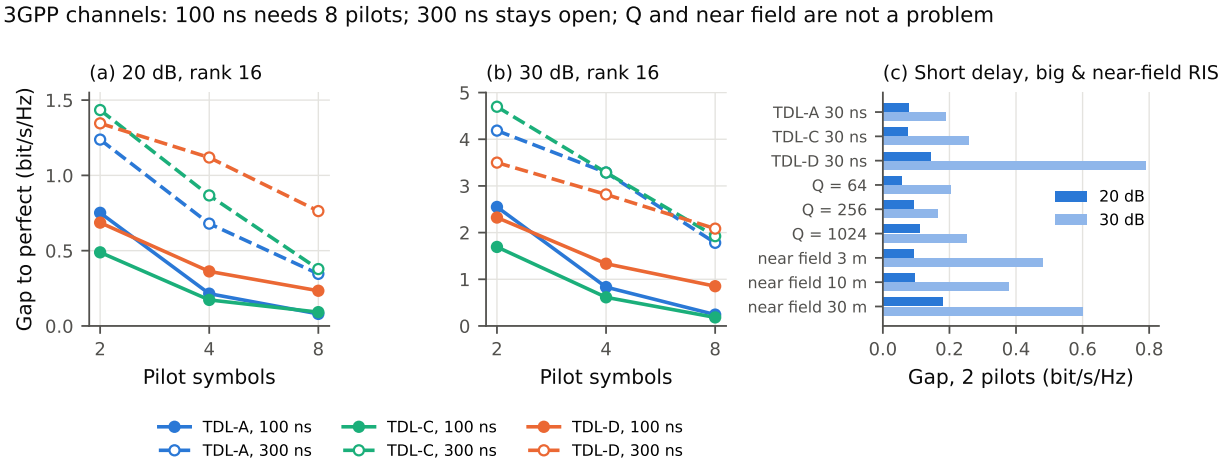


Fig. 11. 3GPP channels: gap versus pilot symbols at 100 and 300 ns; short delay spread, large and near-field surfaces.

J. Timing knowledge and blind offset recovery

Sensitivity. With the nominal timing, an unknown common offset of 0.02 slot (83 ns) costs 4.3/11.3 dB at 20/30 dB, and 0.2 slot costs 15.1/25.3 dB (Fig. 13). The receiver must therefore know or estimate the offset.

Physical timing. Proposition 3 was checked numerically with bulk delays $\tau_b \sim \mathcal{U}(50, 300)$ ns and $\tau_u \sim \mathcal{U}(30, 300)$ ns on every path and synchronization to the first cascaded arrival: both identities hold to $7 \cdot 10^{-15}$ (30 channels). With the RIS locked to the incident signal, the receiver with the nominal timing is within $+0.001 \pm 0.007$ (20 dB) and $+0.040 \pm 0.082$ bit/s/Hz (30 dB) of the bulk-delay-free result, also with 1 ns rms switching jitter ($+0.086 \pm 0.062$ at 30 dB). With the RIS locked to the BS clock, ignoring the

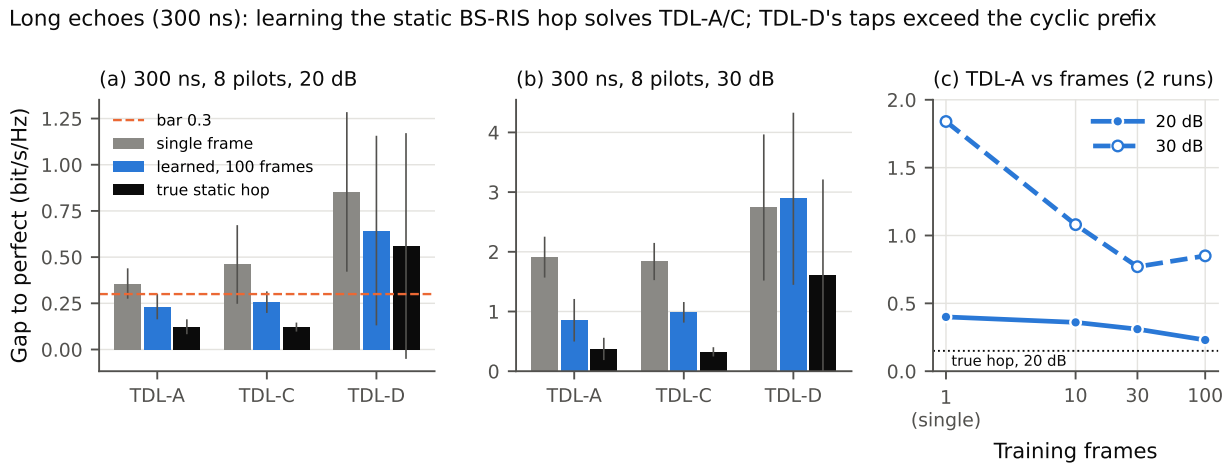


Fig. 12. Learning the static BS–RIS side at 300 ns delay spread.

offset costs 1.62/4.23 bit/s/Hz, while Algorithm 2 is within $+0.011 \pm 0.019$ and $+0.033 \pm 0.066$ bit/s/Hz of a receiver that knows the offset (paired differences, t-based 95% intervals; Table XXII).

TABLE XXII

PHYSICAL TIMING (30 CHANNELS, PAIRED DIFFERENCES WITH T-BASED 95% CIS, BIT/S/HZ).

	20 dB	30 dB
RIS locked to incident signal: nominal timing − no bulk delay	$+0.001 \pm 0.007$	$+0.040 \pm 0.082$
same, 1 ns rms per-edge jitter	$+0.015 \pm 0.016$	$+0.086 \pm 0.062$
RIS locked to BS clock: offset ignored, gap	1.62	4.23
evidence search − known offset	$+0.011 \pm 0.019$	$+0.033 \pm 0.066$
Sync error ± 150 ns, evidence with default window − reference	$+0.22 \pm 0.18$	$+1.16 \pm 0.67$
fixed window margin $[-150\,\text{ns}, \tau_{\max}]$	$+0.21 \pm 0.06$	$+0.79 \pm 0.23$
window start chosen by evidence (Alg. 2)	$\mathbf{+0.05 \pm 0.05}$	$\mathbf{+0.05 \pm 0.08}$

Delay reference. A delay window that starts after the synchronization reference fails completely: with $[100, 400]$ ns, the loss against known timing is 6.68 bit/s/Hz at 30 dB, while $[-100, 200]$ ns and $\tau_{\max} \times 0.5$ or $\times 1.5$ stay within 0.3 dB (a pre-registered test that failed for the first variant). The same failure appears when the receiver synchronizes late. With a synchronization error uniform in ± 150 ns, letting the evidence also choose the window start (Algorithm 2) brings the loss to $+0.05$ bit/s/Hz at both SNRs, whereas a fixed margin does not (Table XXII; fresh channels, criterion fixed before the run).

Residual versus evidence. Earlier residual-based recovery, which scanned a 0.025-slot grid with a reduced 8-iteration estimator and refined only locally, failed at 30 dB (rms error 0.021 slot, loss 4.80 dB against 0.41 dB known). That failure was partly a property of the procedure, not of the criterion: with the full estimator on a 0.01-slot grid, the in-sample residual is not significantly worse than the evidence ($+0.021 \pm 0.032$ and $+0.092 \pm 0.140$ bit/s/Hz at 20/30 dB; one of 30 channels with an error above 0.01 slot, none for the evidence). The overfitting explanation of the Remark was tested directly with two held-out pilot symbols: the held-out residual and the evidence select the same offset (within 0.005 slot) in 93% of channels and give the same rate (-0.001 ± 0.029 at 30 dB). [TODO: insert F2 regimes: coarse procedure, $N_p = 1$, TDL-C 100 ns, wall-clock.] The evidence is therefore preferred for its cost (no low-rank estimation per candidate) and smoothness, not because it is the only criterion that works.

A failed idea. Since an offset is a delay, widening the delay window by a guard should absorb it without a search. It does not: for offsets up to 83 ns the guard window loses 2.32 dB against 1.76 dB without it (0.39 known); for offsets up to 208 ns it improves 12.3 to 9.2 dB, far from 0.28. A wider window spreads the prior over more dimensions; the shifted responses were representable in the plain basis already.

Jitter. Random jitter of individual switching edges changes the pattern's shape, not only its delay, so it is not absorbed: it must stay below about 20 ns rms for 20 dB operation and 4 ns for 30 dB (loss at 4/8/21 ns: 0.24/0.40/0.97 dB at 20 dB, 0.94/2.05/5.91 dB at 30 dB); at 1 ns the loss is $+0.09 \pm 0.06$ bit/s/Hz at 30 dB.

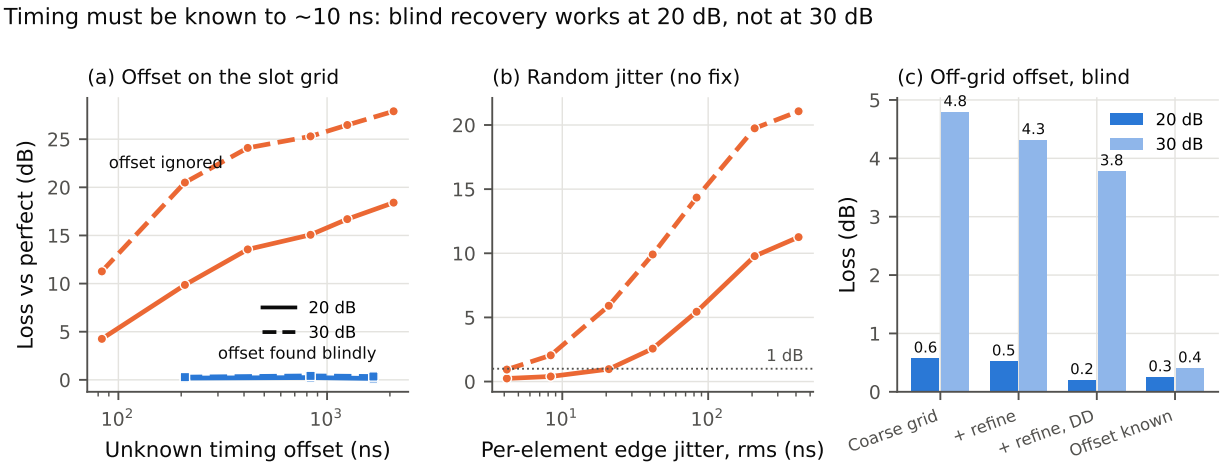


Fig. 13. Timing: offset sensitivity, jitter, residual-based off-grid recovery (coarse procedure).

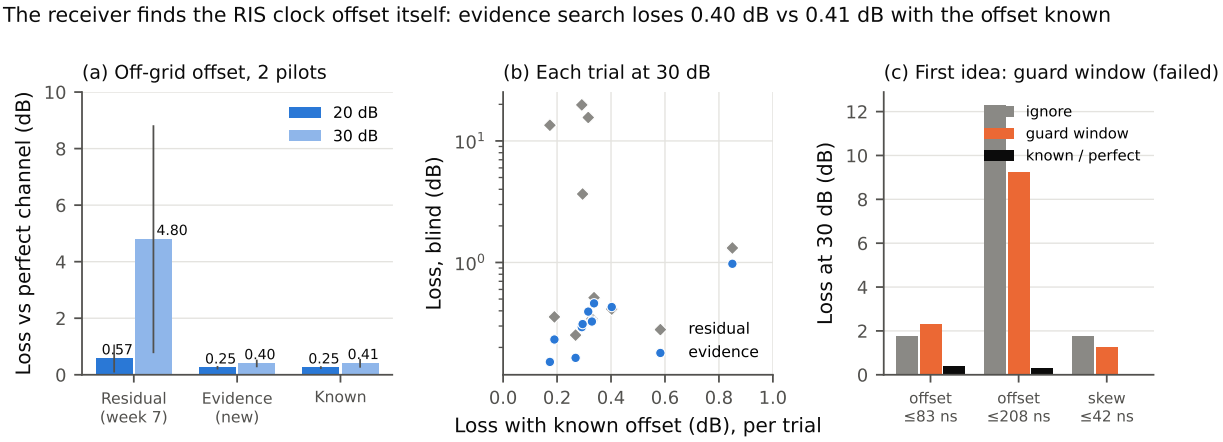


Fig. 14. Evidence-based blind offset recovery (free offset stress test); per-channel losses; the failed guard-window idea.

K. *Hardware impairments*

Element spread. Random resonance spread of 5–20 MHz between elements, with $\pm 5\%$ loss spread, costs 0.20–0.28 / 0.28–0.46 dB at 20/30 dB: the timing-only estimator never assumed identical elements.

Switching transients. A linear ramp between states with rise time ρ (fraction of a slot) costs up to 3.9/11.7 dB if ignored. Building the kernels from a ramped waveform with ρ calibrated once, or chosen blindly by the smallest pilot residual on a grid, brings this to 0.21–0.43 / 0.36–0.61 dB for $\rho = 0.05$–0.3; the blind estimate is within 0.03 of the true rise time.

Realistic rise times. For rise times of 2–42 ns (0.0005–0.01 slot), as fast switches achieve, the nominal instantaneous kernels lose nothing against calibrated ones ($\leq$ 0.011 bit/s/Hz at 30 dB); at 83 ns the uncalibrated receiver loses 0.14 bit/s/Hz at 30 dB and the calibrated one does not. Since emission limits favour slower edges (below), the two requirements combine into a rule: slow the edges to meet the emission limit and let the receiver fit the rise time.

Carrier frequency offset. A residual CFO of 0.5/1/2/5% of the spacing costs 0.26/0.32/0.52/1.70 dB at 20 dB and 0.72/1.00/2.45/7.50 dB at 30 dB: residual CFO must stay below about 1%, as standard OFDM synchronization provides.

Phase noise. With a continuous Wiener phase over pilots and data and common-phase removal per symbol, the fair reference is a genie that knows the channel but not the phase noise. Our receiver adds 0.37, 0.52, 0.90 and 1.20 dB to that reference at 20 dB for 10, 100, 1000 and 10 000 Hz linewidth (0.42, 0.99, 1.37, 1.38 dB at 30 dB). The pre-registered criterion (within 0.5 dB at $\leq$ 1 kHz, 20 dB) failed at 1 kHz and was borderline at 100 Hz: the pilots are corrupted too. Most of the loss, however, is the

generic phase-noise ICI that any receiver suffers (the genie itself falls from 6.63 to 4.72 bit/s/Hz at 1 kHz). Requirement: linewidth below about 100 Hz at 30 kHz spacing, or standard phase tracking (Fig. 15).

Out-of-band emission. Switching spreads energy beyond the occupied band. For full-band loading (3276 subcarriers, 100 MHz channel spacing, $D = 0.75$, $K = 8$), the ratio of in-band to adjacent-channel power of the reflected signal is 36.8 dB with ideal edges, below the 45 dB that 3GPP requires of a base station [34]; a passive RIS has no requirement of its own, but this matters for deployment. A finite rise time fixes it: 42.3, 52.7 and 60.6 dB for rise times of 0.01, 0.05 and 0.1 slot (42, 208, 417 ns), and the second adjacent channel is above 47 dB in all cases (analytic, from the exact harmonic coefficients). Our receiver handles rise times by building its kernels from the calibrated or blindly fitted ramp (above).

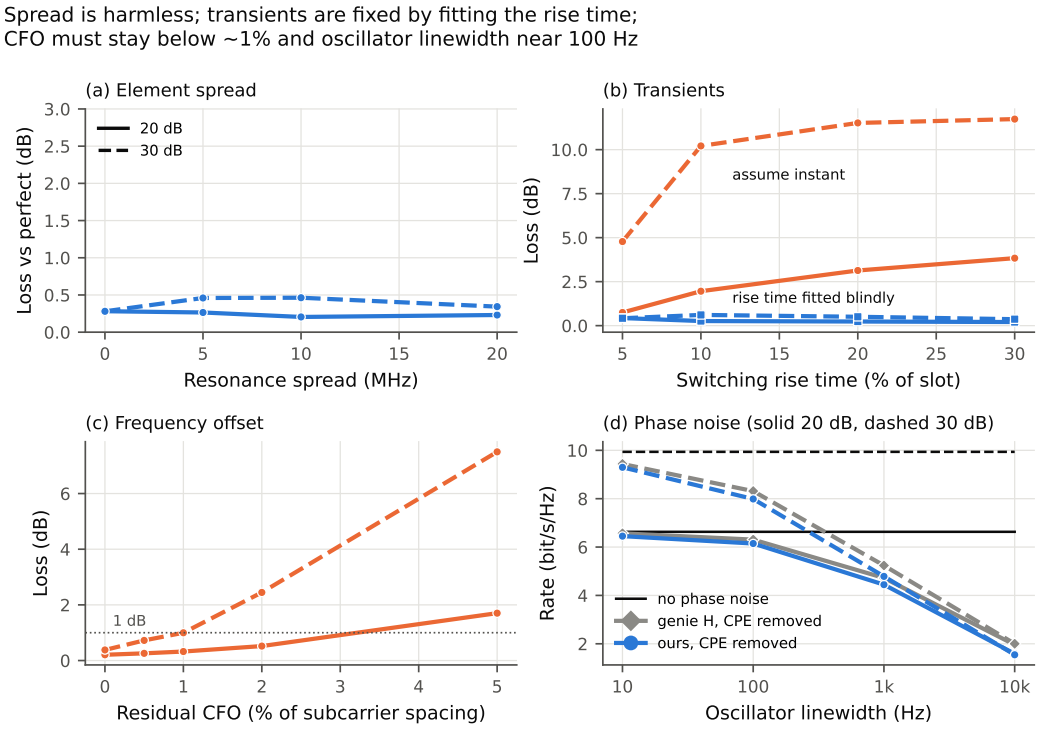


Fig. 15. Element spread, switching transients, CFO and phase noise.

L. Direct path, multiple panels and interference

Unblocked direct path. A direct path at -20 to $+10$ dB relative to the RIS path needs no change to the estimator, by Corollary 6(iii): at 0 dB, 5.28/8.13 against 5.39/8.49 with perfect knowledge (an explicit direct-path term gives the same numbers). The one-tap receiver improves with a strong direct path (2.9–3.1 bit/s/Hz at $+10$ dB) but stays far below.

Two panels. Two TM-RIS panels with independent channels and different patterns ($G = 2$): 5.54/8.67 against 5.64/8.86. A receiver that models only one panel collapses to 1.15: the receiver must know the timing of every panel (A1).

Interference. An unknown OFDM interferer through its own channel, treated as noise, at SIR 20/10/0 dB: within 0.03–0.17 bit/s/Hz of perfect knowledge at every SIR; interference limits both equally (Fig. 16).

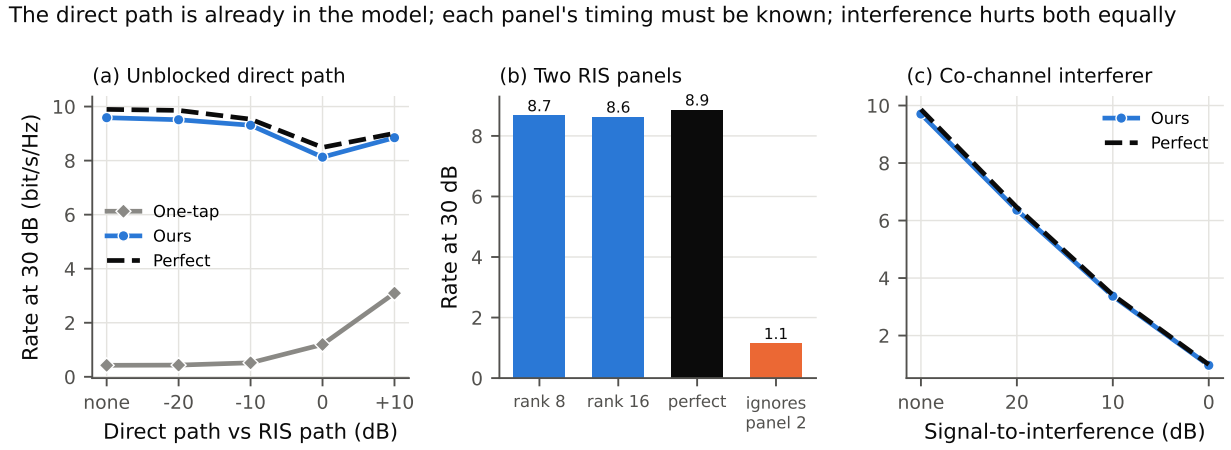


Fig. 16. Unblocked direct path, two panels, co-channel interference.

M. Pilot design

With the same number of pilot resource elements (512) and energy: two full pilot symbols give 6.49/9.71; comb pilots on every second subcarrier over four symbols with the other resource elements *empty*, as NR DMRS allows, give 6.52/9.79; every fourth subcarrier over eight symbols, 6.51/9.77. With *data* in the gaps, the switching leaks data into the pilot bins and the rate falls to 4.79/5.10; one DD pass recovers 6.36/8.82. Design rule: comb pilots work if their neighbours are empty.

N. Mobility, coherence time and pilot overhead

Coherence. Only the RIS–user hop moves. The coherence time $T_c = 0.423/f_D$ with $f_D = vf_c/c$ is 42.4 ms (1188 symbols) at 3 km/h, 4.24 ms (119 symbols) at 30 km/h and 1.06 ms (30 symbols) at 120 km/h. Doppler mixing within a symbol is -60, -47 and -34 dB relative to the signal at these speeds (0, 0.008 and 0.19 bit/s/Hz at 30 dB), so the main effect is the ageing of the estimate. With one estimate reused (per-subcarrier gain metric), the gap at 20 dB stays below 0.1 bit/s/Hz for more than 300 symbols at 3 km/h, reaches 0.26 after 40 symbols at 30 km/h (20 dB), and 0.32 after 10 symbols at 120 km/h (Fig. 17).

Two failed tracking receivers. Tracking $\boldsymbol{U}$ symbol by symbol from decisions, with the input side learned, made things worse than reusing the pilot estimate (average gap over 40 symbols at 20 dB: 0.52 against 0.28); a proper Kalman filter with propagated covariance also failed (0.46). Wrong decisions accumulate after about ten symbols, and at 30 dB even the first symbol after the pilots has a gap of 0.44–0.52 for every receiver, because the in-symbol Doppler mixing at 120 km/h is not in our model. Fast users therefore need frequent pilots, which the overhead analysis quantifies.

Net throughput. Table XXIII and Fig. 18 give the net rate after pilot overhead with the refresh interval and pilot count optimized per case. After paying for its pilots, the proposed receiver beats the best CE-BEM in every one of the 24 cases, by 1.1–4.8 bit/s/Hz (median 2.7). On the 3GPP channels at 20 dB it needs 5–7% overhead at 3 km/h, 5–10% at 30 km/h and 12% at 120 km/h (one pilot symbol every eight). Slow channels with long echoes prefer several pilot symbols per refresh (four, on TDL at walking speed); fast channels prefer one pilot symbol refreshed often. The six-path model needs less overhead at speed (1–3% at 20 dB) because its dominant line-of-sight path makes the Doppler mostly a common phase, which tracking removes; the non-line-of-sight TDL numbers are the realistic ones. Without any phase tracking, 120 km/h requires about 25% overhead.

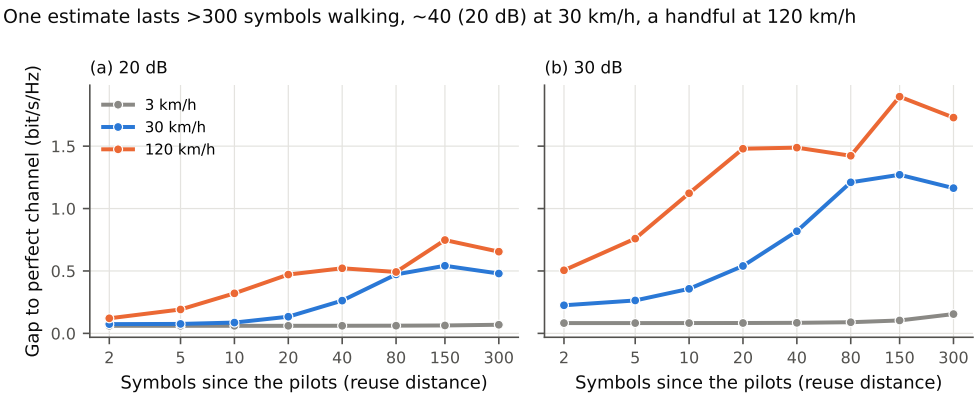


Fig. 17. Gap versus symbols since the pilots at 3, 30 and 120 km/h (estimate reused).

O. Does time modulation pay off for the user? Motivation and secrecy

Table XXIV compares, at fixed transmit power and noise (the noise is set so that an optimized static RIS has the reference SNR), the rate of the legitimate user with no RIS (direct path at -10 dB), a static 1-bit RIS with co-phased or random states, and a TM-RIS with a pattern designed for the user (per-element polarities co-phasing the $h = 0$ harmonic, random cyclic shifts for scrambling; 8–11 distinct patterns, so four pilot symbols are used), and the rate of an eavesdropper at another position (30 channels).

TABLE XXIII

NET THROUGHPUT (BIT/S/HZ) AFTER PILOT OVERHEAD, REFRESH INTERVAL AND PILOT COUNT OPTIMIZED PER CASE; IN BRACKETS: PILOT SYMBOLS / BLOCK LENGTH. OURS WITH COMMON-PHASE TRACKING (MAIN), ITS RANGE FROM NO TRACKING TO A PER-SUBCARRIER GENIE, THE BEST CE-BEM (BEST HARMONIC RANGE AND PILOT COUNT, SAME TRACKING), ONE-TAP, AND PERFECT KNOWLEDGE WITHOUT PILOTS. SHARED-PATH: 8 CHANNELS; TDL: 4 CHANNELS PER POINT. CE-BEM ON TDL IS LIMITED TO $|h| \leq 40$.

Channel	km/h	SNR	ours	ours range	CE-BEM	one-tap	perfect
Shared-path, 30 ns	3	20	6.28 (2/120)	5.98–6.33	5.14 (32/320)	0.38	6.49
Shared-path, 30 ns	3	30	9.29 (2/80)	8.43–9.47	6.87 (16/240)	0.42	9.79
Shared-path, 30 ns	30	20	6.13 (1/40)	5.38–6.53	4.83 (16/240)	0.39	6.60
Shared-path, 30 ns	30	30	8.93 (1/30)	7.21–9.40	6.02 (16/240)	0.42	9.90
Shared-path, 30 ns	120	20	5.59 (1/160)	3.70–6.23	3.43 (8/320)	0.40	6.49
Shared-path, 30 ns	120	30	7.55 (1/10)	4.40–8.56	2.96 (16/320)	0.43	9.78
TDL-A, 100 ns	3	20	5.05 (4/60)	4.91–5.34	3.44 (16/160)	0.33	5.91
TDL-A, 100 ns	3	30	6.61 (4/40)	6.50–7.03	3.25 (16/120)	0.39	8.94
TDL-A, 100 ns	30	20	4.70 (1/10)	4.47–5.29	2.25 (8/40)	0.34	6.31
TDL-A, 100 ns	30	30	5.89 (2/15)	5.41–6.96	1.93 (16/60)	0.40	9.54
TDL-A, 100 ns	120	20	4.26 (1/8)	3.65–4.97	1.53 (16/240)	0.34	6.28
TDL-A, 100 ns	120	30	5.09 (1/6)	4.34–6.08	1.24 (16/240)	0.38	9.51
TDL-C, 100 ns	3	20	5.71 (4/80)	5.55–6.02	3.70 (16/240)	0.38	6.34
TDL-C, 100 ns	3	30	8.05 (4/60)	7.61–8.60	3.33 (16/240)	0.42	9.58
TDL-C, 100 ns	30	20	4.81 (1/10)	4.69–5.20	2.10 (8/40)	0.33	5.95
TDL-C, 100 ns	30	30	6.03 (1/8)	5.80–6.73	1.73 (8/30)	0.37	9.13
TDL-C, 100 ns	120	20	4.34 (1/8)	3.65–4.95	1.66 (16/320)	0.33	6.07
TDL-C, 100 ns	120	30	5.37 (1/6)	4.39–6.39	1.39 (16/320)	0.40	9.30
TDL-D, 100 ns	3	20	5.58 (4/80)	5.22–5.77	3.63 (16/160)	0.39	6.25
TDL-D, 100 ns	3	30	8.06 (2/60)	7.46–8.51	3.26 (16/160)	0.40	9.51
TDL-D, 100 ns	30	20	5.06 (1/20)	4.53–5.60	2.88 (32/320)	0.37	6.49
TDL-D, 100 ns	30	30	7.06 (1/15)	5.96–7.73	2.55 (16/320)	0.42	9.75
TDL-D, 100 ns	120	20	3.91 (1/8)	3.10–4.74	1.64 (16/160)	0.32	6.13
TDL-D, 100 ns	120	30	4.94 (1/8)	3.66–5.51	1.36 (16/160)	0.34	9.32

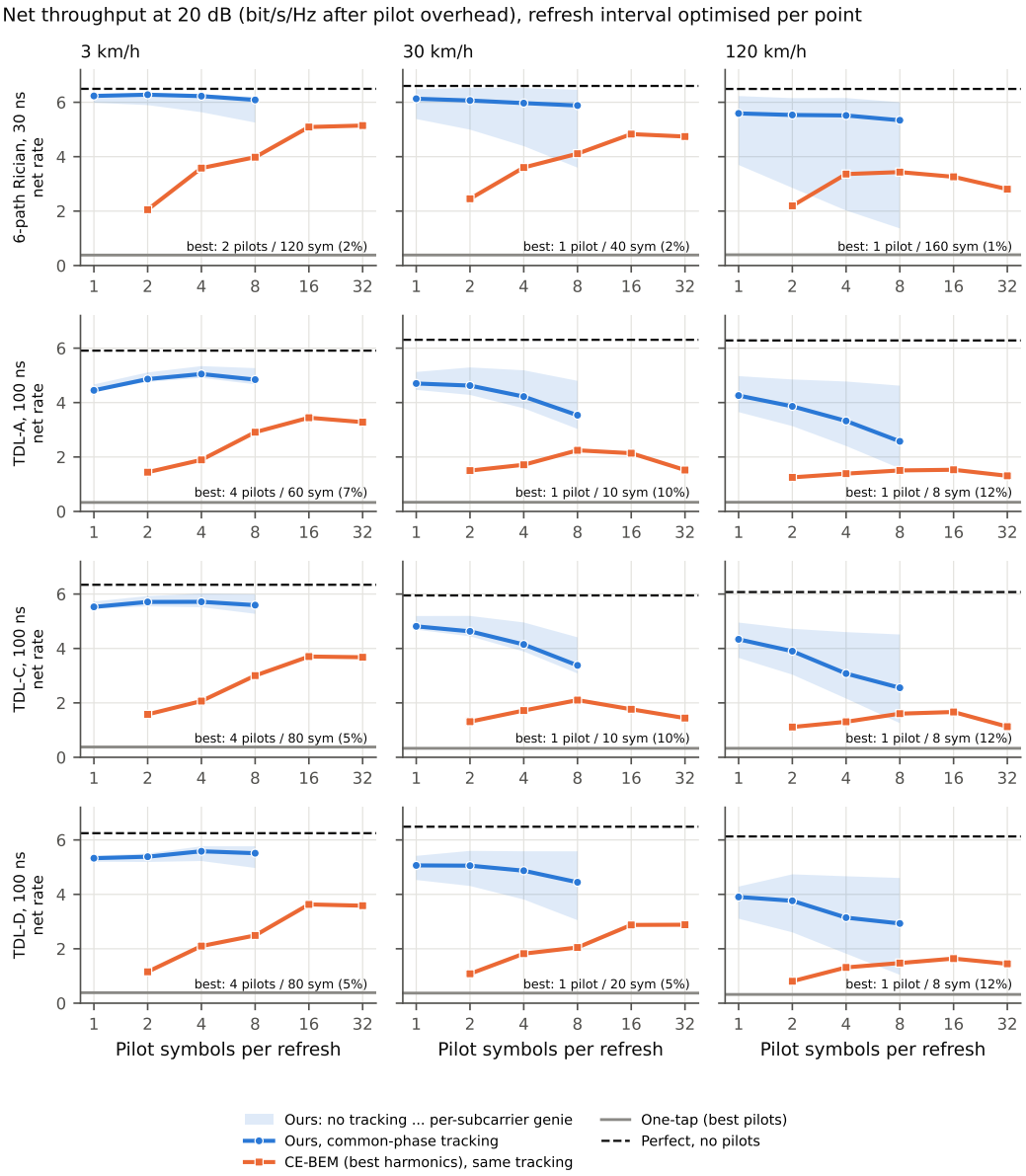


Fig. 18. Net throughput at 20 dB after pilot overhead versus pilot symbols per refresh.

Three conclusions follow. (i) The proposed receiver recovers most of what time modulation takes from the legitimate user: $+2.1$ to $+2.3$ bit/s/Hz over one-tap at 20 dB ($+4.9$ to $+5.2$ at 30 dB, $+0.3$ to $+0.7$ at 10 dB), within 0.1–0.2 of perfect knowledge. (ii) For a single user, time modulation does not pay off: the co-phased static RIS gives 1.3–3.7 bit/s/Hz more at 20 dB. Time modulation is justified only by its other functions, and our contribution is to make its cost to the legitimate user small. (iii) The secrecy of this time-modulated design is limited even without pattern knowledge: an eavesdropper with a generic

TABLE XXIV

RATE AT 20 DB (BIT/S/HZ) AT FIXED TRANSMIT POWER; TM-RIS WITH A PATTERN DESIGNED FOR THE USER, $K = 8$.

	$D = 0.625$	$D = 0.75$	$D = 0.875$
No RIS (direct path, -10 dB)	3.46	3.45	3.46
Static RIS, random states	3.09	3.40	3.29
Static RIS, co-phased states	6.66	6.66	6.66
TM-RIS, one-tap	0.68	1.98	3.15
TM-RIS, proposed ($N_p = 4$)	2.98	4.06	5.35
TM-RIS, perfect CSI	3.08	4.18	5.50
Eavesdropper, one-tap	0.08	0.47	0.71
Eavesdropper, CE-BEM $\lvert h\rvert \leq 40$, no pattern	2.04	1.99	1.79
Eavesdropper, proposed, *with* the pattern	2.76	2.71	2.47

CE-BEM estimator and the same four pilots obtains 1.8–2.2 bit/s/Hz; with the pattern and our receiver it obtains 2.5–3.0, and the secrecy rate falls to 0.5–2.9 bit/s/Hz. If the pattern is broadcast as control information (A1), part of the security is lost; keeping it secret, or designing patterns for secrecy against model-aware receivers, is a question for TM-RIS security design rather than for the receiver. Varying K (4, 8, 16) at $D = 0.75$ changes these numbers by less than 0.3.

P. Measured cost

Updates for a moving user ($M = 256$, 30 km/h, refresh every 20 symbols, 8 channels; Table XXV). Warm start with two iterations needs 12$\times$ fewer operations than estimation from scratch for 0.03 bit/s/Hz more gap. Freezing $\boldsymbol{V}$ is about 100$\times$ cheaper and as good at 20 dB, but costs 0.23 bit/s/Hz more at 30 dB, because the rank-8 factorization does not split the static and moving hops exactly; Algorithm 3 with $r = Q$ addresses this split. At the full band, a warm update takes 1.4 s against 12.5 s from scratch in MATLAB (8.6$\times$), with the same SER (0.399 vs 0.398; both estimates are 20 symbols old at 30 km/h, which sets the absolute SER, see the caveat on tracking above).

Equalizer. The pre-registered wall-clock comparison *failed* with the element-by-element FFT operator: at $M = 3276$ CG with 25 iterations took 18.7 s for 12 symbols against 5.6 s for the dense solve, with the same SER (0.065), because many small FFT calls are slow in MATLAB. With the batched operator of Algorithm 5 and the stopping rule (a new pre-registered criterion), CG took 1.12 s (5 iterations) against 2.95 s dense at $M = 3276$, 2.6$\times$ faster with the same SER, and 1.85 s at $M = 6552$, where the dense

TABLE XXV

UPDATE VARIANTS FOR A MOVING USER ($M = 256$): GAP AT 20/30 DB, OPERATIONS, MATLAB TIME.

Update	gap	operations	saving	time
From scratch, 20 it., $r = 8$	0.07 / 0.17	$1.2 \cdot 10^9$	1×	307 ms
From scratch, $r = 6$	0.07 / 0.16	$7.2 \cdot 10^8$	1.7×	215 ms
Warm start, 2 it. (Alg. 4)	0.10 / 0.20	$1.0 \cdot 10^8$	12×	31 ms
$\boldsymbol{V}$ frozen, 1 it.	0.10 / 0.40	$1.1 \cdot 10^7$	105×	8 ms
$\boldsymbol{V}$ frozen, 1 pilot symbol	0.13 / 0.49	$6.0 \cdot 10^6$	197×	5 ms

solve does not fit in memory (687 MB per matrix). At $M = 2048$ the two are equal in time and the stopping rule left CG's SER 1.18× that of the dense solve, which suggests a tighter tolerance there. The dense solve's time varied by 2× between runs with machine load; comparisons are within a run (Fig. 19).

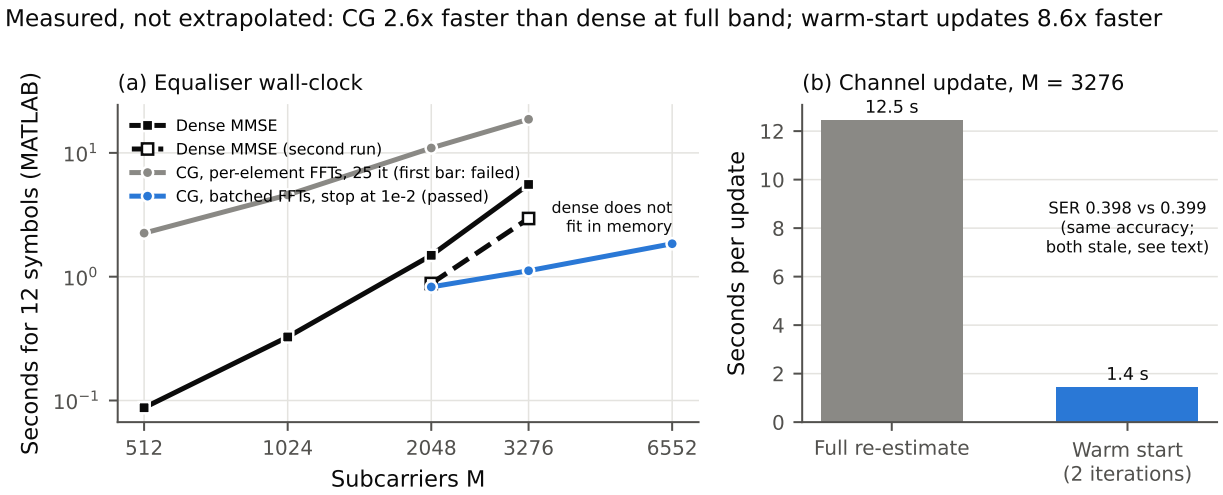


Fig. 19. Measured equalizer time versus M and full-band update time.

VIII. TRADE-OFFS, LIMITATIONS AND SCOPE

A. *Trade-offs*

Table XXVI collects the design trade-offs of the receiver, each quantified by an experiment of Section VII.

TABLE XXVI

DESIGN TRADE-OFFS AND THEIR MEASURED CONSEQUENCES.

Choice	Gain	Cost (measured)
Pilot symbols per refresh, N_p	Accuracy: TDL 100 ns gap 0.7 $\rightarrow$ 0.1–0.24 for $N_p = 2 \rightarrow 8$	Overhead N_p/N; for fast users one pilot refreshed often is better (net rate, Table XXIII)
Refresh interval N	Lower overhead	Estimate ages: 120 km/h needs $N \approx 8$ on TDL (12%); without phase tracking $N \approx 4$ (25%)
Delay window $\tau_{\max}$, basis size L	Covers longer echoes (TDL 100 ns needs 1 μs)	More unknowns $\Rightarrow$ more pilots; window beyond the CP is useless (TDL-D, 300 ns)
Rank r	Captures more paths	Little effect between 8 and 32 on TDL; $r = 6$ instead of 8 is free and 1.7$\times$ cheaper
Distinct patterns G (security, scrambling)	Stronger scrambling for others	Unknowns $+Gr^2$: gap 0.10 $\rightarrow$ 1.03 at 30 dB for $G = 1 \rightarrow 16$ with 2 pilots; 4 pilots restore 0.38
Duty cycle D	Away from 0.5: more energy on $h = 0$, better one-tap	Our receiver is indifferent (gap 0.05–0.23 for $D = 1/8..7/8$); the one-tap ceiling $(2D-1)^2/(1-(2D-1)^2)$ changes
Decision-directed passes	28 dB/64-QAM: SER ratio 1.54 $\rightarrow$ 1.14 with one pass	A second pass with 64-QAM raised SER to 0.24 (error propagation); we use at most one
Evidence timing search	Removes the timing requirement; no rate loss vs known timing	55 first-stage evaluations per search, about $1.2 \cdot 10^{12}$ operations at $M = 3276$ (computed, not yet measured); needed only when the offset changes
Two-timescale learning	fewer per-frame unknowns (gain shown only at 300 ns, a CP-violating case)	Needs $\sim$100 frames sharing the BS–RIS hop; stale if the BS or RIS moves
Warm start	12$\times$ fewer operations (8.6$\times$ faster at full band)	+0.03 bit/s/Hz gap
Freezing $\boldsymbol{V}$ in updates	$\sim$100$\times$ fewer operations	+0.23 bit/s/Hz at 30 dB unless the input side is learned with $r = Q$
CG tolerance / iterations	Linear cost in M; fits where dense does not	Loose tolerance (10^{-2}) left SER 1.18$\times$ dense at $M = 2048$; iterations grow with bandwidth
Switching rise time	Lower emission (ACLR 36.8 $\rightarrow$ 52.7 dB for 0 $\rightarrow$ 0.05 slot)	Shrinks our advantage over truncated estimators (CE-BEM 4.7 $\rightarrow$ 8.2 at 0.3 slot); calibrated kernels needed above $\sim$40 ns
Comb pilots	Same accuracy as full pilot symbols, NR-compatible	Neighbouring resource elements must be empty (data next to pilots: 9.79 $\rightarrow$ 5.10 at 30 dB)

B. Limitations and scope

- *Simulation.* All results are simulated with one published element model; mutual coupling between elements is ignored, as in the anchor model. Over-the-air validation is future work.
- *Pattern knowledge.* The receiver must know every panel's switching pattern (A1); with a wrong pattern for one of two panels the rate collapses from 5.5 to 1.2 bit/s/Hz.
- *Delay spread beyond the prefix.* Channels whose taps exceed the cyclic prefix (TDL-D at 300 ns) are out of scope of any prefix-limited window; 300 ns TDL-A/C need the static side learned over $\sim$100 frames.
- *High speed.* At 120 km/h a pilot symbol every $\sim$8 symbols is needed on TDL channels (12% overhead at 20 dB). Two tracking receivers did not improve on this, and at 30 dB the in-symbol Doppler mixing, not modelled by our estimator, limits every receiver. A Doppler-aware extension of the model is future work.
- *Mobility metric.* All mobility numbers assume common-phase tracking per symbol, as in 5G NR; without it pilots are needed about twice as often.
- *Hardware.* Switching jitter must stay below about 4 ns at 30 dB, residual CFO below about 1% of the spacing, and oscillator linewidth near 100 Hz or tracked; transients are handled by fitting the rise time.
- *Low SNR.* At 10 dB the gain over the one-tap receiver for a user the pattern is designed for is modest (+0.6 bit/s/Hz, about 2 dB): noise, not coupling, dominates there.
- *Motivation and secrecy.* For a single user an optimized static RIS gives more rate than a TM-RIS, also with our receiver (Section VII-O); our receiver reduces the cost of time modulation, it does not make it profitable. An eavesdropper that learns the pattern can run the same receiver (secrecy rate 0.5–2.9 instead of 1.0–3.6 bit/s/Hz at 20 dB).
- *Rank.* With $Q = 16$ and $r = 8$ (or $r = Q$ in Algorithm 3) the low-rank property is weak; it matters for large surfaces, where it is tested only up to $Q = 1024$ with narrowband array phases.
- *Equalizer speed.* The measured CG advantage counts the dense factorization against 12 symbols; with more symbols per estimate the dense factorization is amortized, and the break-even number of symbols has not been measured.
- *Statistics.* Some points use 4–8 channels (300 ns, TDL overhead) and the full-band 28 dB result 10 channels; their confidence intervals are wider than those of the 20–30-channel results.
- *Learning-based estimators* were not compared (Section VI).

IX. Conclusion

The harmonic coupling of a time-modulated RIS is not unknown interference: it is the product of a switching timing that the network sets and the receiver knows, and a propagation-plus-dispersion part that it does not. Grouping the harmonic expansion by switching state makes this separation exact. Built on it, a receiver needs no element model; estimates the coupling from two to eight pilot symbols in a low-dimensional shared delay subspace; finds the RIS clock offset by itself, once one recognizes that the offset is an opposite delay of the two hops and uses the evidence rather than the fit; learns the static base-station side once; updates cheaply; and equalizes without forming the coupling matrix. Against the strongest baselines we could build, including an ideal-element estimator with our prior and robust equalizers for all, it is ahead on every channel and pilot count, exceeds with two pilot symbols a full-harmonic basis expansion given 32, decodes where the conventional receiver never does, and after paying for its pilots keeps 3.9–5.7 bit/s/Hz on 3GPP channels from walking speed to 120 km/h. The limits found, delay spreads beyond the cyclic prefix, high-speed users at high SNR, and the need to know every panel's pattern, point to the natural next steps: a Doppler-aware coupling model, extended-prefix operation, and over-the-air validation with a real time-modulated surface.

Appendix A
Energy on the original subcarrier

For ideal elements $\Gamma_1 = +1$, $\Gamma_2 = -1$ and a common pattern, the switched response is $\gamma(t) = \mathbb{1}_1(t) - \mathbb{1}_2(t) \in \{\pm 1\}$ with Fourier coefficients $b^{[h]} = t_1[h] - t_2[h]$. Then $b^{[0]} = D - (1-D) = 2D-1$, and by Parseval $\sum_h |b^{[h]}|^2 = \frac{1}{T_u}\int_0^{T_u} |\gamma(t)|^2 dt = 1$. A one-tap receiver uses only $b^{[0]}$: the useful fraction is $(2D-1)^2$ and the rest, $1-(2D-1)^2$, reaches the subcarrier from its neighbours as interference, so SIR $\le (2D-1)^2/(1-(2D-1)^2)$; at $D = 0.75$, $0.25/0.75 = -4.77$ dB, and at $D = 0.875$, $0.5625/0.4375 = +1.09$ dB. Joint equalization collects the full unit energy, a gain of $1/(2D-1)^2$: 12.0, 6.02 and 2.5 dB at $D = 0.625$, 0.75 and 0.875. The numerical sum of (3) over $|h| \le 20000$ gives 0.99998, confirming the normalization used for the tail fractions of Corollary 3.

Appendix B
Evaluating the evidence

Let $\boldsymbol{\Phi} = \boldsymbol{\Phi}_d$, $n = MN_p$ and $\boldsymbol{P}^{1/2}$ diagonal. With $\boldsymbol{C} = \sigma^2\boldsymbol{I} + \boldsymbol{\Phi}\boldsymbol{P}\boldsymbol{\Phi}^{\mathsf{H}}$ and $\boldsymbol{B} = \boldsymbol{I} + \sigma^{-2}\boldsymbol{P}^{1/2}\boldsymbol{\Phi}^{\mathsf{H}}\boldsymbol{\Phi}\boldsymbol{P}^{1/2} = \boldsymbol{R}^{\mathsf{H}}\boldsymbol{R}$, the matrix determinant lemma gives $\det \boldsymbol{C} = \sigma^{2n}\det \boldsymbol{B}$, and the Woodbury identity gives $\boldsymbol{C}^{-1} = \sigma^{-2}\big(\boldsymbol{I} - \sigma^{-2}\boldsymbol{\Phi}\boldsymbol{P}^{1/2}\boldsymbol{B}^{-1}\boldsymbol{P}^{1/2}\boldsymbol{\Phi}^{\mathsf{H}}\big)$. Hence

$$\log p(\boldsymbol{y}|d) = -\frac{\|\boldsymbol{y}\|^2}{\sigma^2} + \frac{\|\boldsymbol{R}^{-\mathsf{H}}\boldsymbol{P}^{1/2}\boldsymbol{\Phi}^{\mathsf{H}}\boldsymbol{y}\|^2}{\sigma^4} - 2\sum_i \log R_{ii} + \text{const}, \tag{14}$$

which needs one Cholesky factorization of a $2L^2 \times 2L^2$ matrix per candidate. The scale ς in $\boldsymbol{P}$ is set per candidate by moment matching, $\varsigma = (\|\boldsymbol{y}\|^2/n - \sigma^2)/(\sum_i \|\boldsymbol{\phi}_i\|^2 p_i/n)$, as in the estimator.

APPENDIX C
DERIVED QUANTITIES

Coherence time. $f_D = v f_c/c$ with $f_c = 3.594\,\text{GHz}$: 9.98, 99.8 and 399.3 Hz at 3, 30 and 120 km/h; $T_c = 0.423/f_D$ [32]: 42.4, 4.24 and 1.06 ms; divided by $T_s = 1/\Delta f + T_{\text{cp}} = 35.67\,\mu\text{s}$: 1188, 119 and 30 symbols.

Unknown counts (Table III). $L = \lceil M\Delta f \tau_{\max} \rceil + 10$: $\lceil 256 \cdot 30\,\text{kHz} \cdot 0.3\,\mu\text{s} \rceil = \lceil 2.30 \rceil = 3$ is the bandwidth–delay product, and $L = 16$ was fixed in calibration; at the full band $\lceil 3276 \cdot 30\,\text{kHz} \cdot 0.2\,\mu\text{s} \rceil + 10 = \lceil 19.66 \rceil + 10 = 30$. Ours: $3Lr + Gr^2 = 3 \cdot 16 \cdot 8 + 64 = 448$ ($G = 1$), $384 + 16 \cdot 64 = 1408$ ($G = 16$), $3 \cdot 30 \cdot 8 + 64 = 784$ (full band). CE-BEM: $(2H_b + 1)L_2$ with $L_2 = \lceil 256 \cdot 30\,\text{kHz} \cdot 0.6\,\mu\text{s} \rceil + 12 = 17$: 425, 1377, 4369 and 8687 for $H_b = 12$, 40, 128 and 255.

Operation counts (Table IV) evaluate (12) with $N_p = 2$, $r = 8$, 20 iterations and FFT length $N_{\text{f}} = 2^{\lceil \log_2 4M \rceil}$. Dense storage is $16M^2$ bytes (complex double): 172 MB at $M = 3276$ and 687 MB at $M = 6552$.

Losses in dB are rate gaps divided by the local slope of the perfect-knowledge rate, 0.31 bit/s/Hz per dB at 20 dB and 0.33 at 30 dB, measured on the same channels.

Coded crossing. Interpolating $\log_{10}$BLER linearly in SNR between 8 and 10 dB: proposed $8 + 2 \cdot \frac{\log_{10} 0.1 - \log_{10} 0.275}{\log_{10} 0.025 - \log_{10} 0.275} = 8.84\,\text{dB}$; perfect knowledge $8 + 2 \cdot \frac{\log_{10} 0.1 - \log_{10} 0.208}{\log_{10} 0.0083 - \log_{10} 0.208} = 8.46\,\text{dB}$.

APPENDIX D
REPRODUCIBILITY

All code is base MATLAB (R2026a); figures are drawn by one Python script from the saved results. Table XXVII maps every result to its script. Pre-registered criteria are recorded with dates in the project log before the corresponding runs.

REFERENCES

[1] I. Iudice, D. Darsena, G. Gelli, and V. Galdi, "Physically consistent modeling of dispersive time-modulated reconfigurable intelligent surfaces for wideband OFDM," arXiv:2609.18360, Sep. 2026.

[2] E. Basar, M. Di Renzo, J. de Rosny, M. Debbah, M.-S. Alouini, and R. Zhang, "Wireless communications through reconfigurable intelligent surfaces," *IEEE Access*, vol. 7, pp. 116753–116773, 2019.

[3] Q. Wu and R. Zhang, "Intelligent reflecting surface enhanced wireless network via joint active and passive beamforming," *IEEE Trans. Wireless Commun.*, vol. 18, no. 11, pp. 5394–5409, Nov. 2019.

[4] L. Zhang *et al.*, "Space-time-coding digital metasurfaces," *Nat. Commun.*, vol. 9, art. 4334, 2018.

[5] [TODO: first author initials] Xu and A. P. Petropulu, "Time-modulated intelligent reflecting surface for waveform security," arXiv:2310.13210, 2023 (ICASSP 2024). [TODO: check]

[6] [TODO: initials] Tao and A. P. Petropulu, "Secure time-modulated IRS via GFlowNets," arXiv:2506.14992, 2025. [TODO: check exact title]

[7] [TODO: initials] Tao, A. P. Petropulu, and H. V. Poor, "Time-modulated IRS for integrated sensing and secure communication," arXiv:2509.05565, 2025. [TODO: check exact title]

[8] [TODO: initials] Tao, [TODO: initials] Xu, and A. P. Petropulu, "How secure is TMA-enabled OFDM directional modulation?," arXiv:2310.08551, *IEEE Trans. Wireless Commun.*, 2025. [TODO: check]

[9] K. Wang, S. Xiang, and A. Petropulu, "Generative control of time-modulated intelligent reflecting surfaces for scalable directional modulation," arXiv:2610.12773, Oct. 2026.

[10] [TODO: initials] Verde, D. Darsena, and V. Galdi, "Rapidly time-varying reconfigurable intelligent surfaces," arXiv:2304.11912, 2023. [TODO: check exact title]

[11] Z. Tang, R. C. Cannizzaro, G. Leus, and P. Banelli, "Pilot-assisted time-varying channel estimation for OFDM systems," *IEEE Trans. Signal Process.*, vol. 55, no. 5, pp. 2226–2238, May 2007.

[12] T. Zemen and C. F. Mecklenbräuker, "Time-variant channel estimation using discrete prolate spheroidal sequences," *IEEE Trans. Signal Process.*, vol. 53, no. 9, pp. 3597–3607, Sep. 2005.

[13] D. Slepian, "Prolate spheroidal wave functions, Fourier analysis, and uncertainty—V: The discrete case," *Bell Syst. Tech. J.*, vol. 57, no. 5, pp. 1371–1430, 1978.

[14] J. A. Tropp and A. C. Gilbert, "Signal recovery from random measurements via orthogonal matching pursuit," *IEEE Trans. Inf. Theory*, vol. 53, no. 12, pp. 4655–4666, Dec. 2007.

[15] P. Schniter, "Low-complexity equalization of OFDM in doubly selective channels," *IEEE Trans. Signal Process.*, vol. 52, no. 4, pp. 1002–1011, Apr. 2004.

[16] L. Rugini, P. Banelli, and G. Leus, "Simple equalization of time-varying channels for OFDM," *IEEE Commun. Lett.*, vol. 9, no. 7, pp. 619–621, Jul. 2005.

[17] T. Hrycak, S. Das, G. Matz, and H. G. Feichtinger, "Low complexity equalization for doubly selective channels modeled by a basis expansion," *IEEE Trans. Signal Process.*, vol. 58, no. 11, pp. 5706–5719, Nov. 2010.

[18] B. Zheng and R. Zhang, "Intelligent reflecting surface-enhanced OFDM: Channel estimation and reflection optimization," *IEEE Wireless Commun. Lett.*, vol. 9, no. 4, pp. 518–522, Apr. 2020.

[19] P. Wang, J. Fang, H. Duan, and H. Li, "Compressed channel estimation for intelligent reflecting surface-assisted millimeter wave systems," *IEEE Signal Process. Lett.*, vol. 27, pp. 905–909, 2020.

[20] C. Hu, L. Dai *et al.*, "Two-timescale channel estimation for reconfigurable intelligent surface aided wireless communications," arXiv:1912.07990, 2019. [TODO: full author list and journal version]

[21] [TODO: authors], "Cooperative reflection and synchronization design for distributed multiple-RIS communications," arXiv:2205.02426, 2022.

[22] H. Chen, R. Long, Y.-C. Liang, and R. Schober, "Blind timing estimation and signal detection for RIS-assisted symbiotic radio with imperfect symbol synchronization," in *Proc. IEEE ICC*, 2024, pp. 4155–4160.

[23] [TODO: authors and exact title], Bayesian model selection for OFDM channel estimation under residual timing offsets, *IEEE Access*, 2019 (University of Glasgow eprint 188514).

[24] [TODO: authors and exact title], multi-model averaging over timing offset and channel length for CFO estimation, *IEEE Trans. Wireless Commun.* (letter), 2010 (HKU repository 10722/124742).

[25] X. Liu *et al.*, "Verification and redesign of OFDM backscatter," in *Proc. USENIX NSDI*, 2021. [TODO: check author list]

[26] [TODO: authors], "Inter-carrier interference reduction technique for backscattered OFDM," 2024. [TODO: venue]

[27] [TODO: authors], "Interference-free backscatter communications for OFDM-based symbiotic radio," arXiv:2408.09192, 2024.

[28] [TODO: authors], "Integrated sensing and communication based on space-time-coding metasurfaces," *Nat. Commun.*, 2025.

[29] [TODO: authors], "Space-time coded differential modulation for reconfigurable intelligent surfaces," arXiv:2508.10244, 2025.

[30] [TODO: authors], "Cyclic short-time varying channel estimation in OFDM power-line communication," [TODO: venue, year].

[31] D. J. C. MacKay, "Bayesian interpolation," *Neural Comput.*, vol. 4, no. 3, pp. 415–447, 1992.

[32] T. S. Rappaport, *Wireless Communications: Principles and Practice*, 2nd ed. Upper Saddle River, NJ, USA: Prentice Hall, 2002.

[33] 3GPP, "Study on channel model for frequencies from 0.5 to 100 GHz," TR 38.901.

[34] 3GPP, "NR; Base Station (BS) radio transmission and reception," TS 38.104.

[35] 3GPP, "NR; Physical channels and modulation," TS 38.211.

TABLE XXVII

RESULTS AND THE SCRIPTS THAT PRODUCED THEM (TM_RIS_OFDM/MATLAB/; RESULTS IN RESULTS/<SCRIPT>.JSON).

Result	Script
Model validation, Fig. 2, 3	`gate1_reproduce`, `td_simulate`
Energy collection, Fig. 4	`gate2_equalization`
Element knowledge, Table VII	`week3_mismatch`, `week3_modelfree`, `week3_modelfree_confirm`
Estimators, Table IX	`week4_bem_compare`
Strong baselines, Tables X, XII	`week7_strong_baselines`, `week6_baselines_pilots`
Designed user, Table XIV	`week3_designed_user`
Full band, Table XVI	`week4_fullband_trials`, `week4_fullband_s2_shift`
Coded, Table XVII	`week6_coded`
Sensitivity, Fig. 10	`week5_param_sweep`
3GPP, near field, Table XVIII	`week6_channels`, `week6_channels_fix`
300 ns, Table XXI	`week8_twoscale_300ns(_b,_c)`
Timing, Table XXII	`week6_timing(_offgrid)`, `week7_timing_refine`, `week8_timing_guard`, `week8_timing_evidence`, `week9_timing_physics`, `week9_timing_landscape`, `week9_prior_sensitivity`, `week9_sync_window`, `week9_resid_vs_evidence`
Model validity, Table VIII	`week9_td_truth`, `week9_mismatch`
Hardware	`week6_hardware(_fix)`, `week6_phasenoise`
Direct path, panels, interference	`week6_directlink`
Mobility, overhead, Table XXIII	`week5_mobility`, `week8_tracking_120(b)`, `week8_pilot_overhead`
Cost, Table XXV, Fig. 19	`week7_complexity_fix`, `week8_fullband_warm`, `week8_eq_walltime(2,3)`